BINARY ELLIS–BRONNIKOV WORMHOLES IN 3+1 NUMERICAL RELATIVITY: MERGERS, FLY-BYS AND GRAVITATIONAL WAVES

NIKITA M. SHIROKOV[1, *] and ILYA NACHEVSKY[2, †]

[1] *Moscow Institute of Physics and Technology, Moscow, Russia*
[2] *Higher School of Economics, Moscow, Russia*

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We present what are, to our knowledge, the first 3D numerical-relativity evolutions of wormhole binaries: pairs of massive Ellis–Bronnikov drainholes held open by a ghost scalar field. Each throat is an unstable fixed point whose e-fold time matches the González–Guzmán–Sarbach linear rate to within 1.9%. Two throats obey a charge-sign rule: like scalar charges repel and opposite charges attract, with a measured attraction-to-repulsion ratio of 1.462 ± 0.022 against the predicted 3/2, so only opposite-signed pairs merge. A head-on collision forms a common MOTS around both still-open throats, born with about their combined area. The horizon then loses 28% of its area as it swallows the phantom field, and by $t = 100$ lies within 1.4% of the Schwarzschild radius of the pair's mass, still shrinking. Fly-by pairs form no horizon and both mouths inflate, the one nearest capture radiating the most gravitational-wave energy measured. A lone collapsing throat sheds its seeded quadrupole as a burst ringing at the Schwarzschild frequency of the black hole it becomes, and every encounter radiates its own burst of gravitational waves and a scalar dipole of comparable energy and negative sign. A throat survives at most a few orbital periods for any plausible perturbation, and the negative-energy dipole would pump rather than drain the orbit, so these binaries have no inspiral.

I. INTRODUCTION

Wheeler's spacetime foam is a sea of handles—wormholes that open and close on the Planck scale [1–3]. Several mechanisms promote them to macroscopic size: inflation stretches a Planck-scale Lorentzian wormhole with the background [4]; bubble and domain-wall cosmologies produce post-inflationary regions joined through wormholes [5, 6, 80, 81]; and a gas of relic wormholes has been studied as a cosmology in its own right [7, 8]. The bubble and wall wormholes are not traversable and pinch off into black holes [6, 80]; none of this work evolves a traversable relic—what one does, and what two of them do to each other. That is the subject of this paper.

Two classical facts govern the dynamics. First, the simplest evolvable traversable wormhole—the Ellis–Bronnikov drainhole held open by a ghost scalar [9–12]—is unstable [18, 55, 76]: depending on the sign of the perturbation, the throat collapses to a black hole or inflates, on a timescale set by the throat radius [13, 17, 64, 75, 77, 79], as 3+1 evolutions of the massless throat confirm nonlinearly [44]. As a quantum field coupled to gravity a ghost destabilises the vacuum unless its effective theory is cut off at low energy with Lorentz violation [62]; we treat it as a classical phenomenological model. Second, Geroch's theorem forbids classical topology change [19, 20]: two wormholes cannot fuse into one, and none can be created classically. Creation is quantum [21], so our evolutions start at $t = 0$ from the newborn state.

Together these facts set selection rules. No astrophysical assembly channel exists: a throat decays faster than dynamical friction or radiation reaction could pair it. A primordial pair born from foam needs no assembly—birth and collision coincide, at a separation comparable to the objects—and its evolution is a race between orbital collision and throat decay, both rates fixed by the classical solution. The scalar adds a second rule: a phantom field repels like charges and attracts opposite ones, so like-signed pairs scatter while opposite-signed pairs fall together; and the sign of a lone throat's perturbation selects collapse—a wormhole-to-primordial-black-hole channel beside the standard ones [22–24]—or inflation. A newborn population thus splits into four fates—merger, scattering, collapse, inflation—plus a fast plunge to a merged core whose end state the grid does not resolve, across birth separation, perturbation and relative sign, the space mapped here (Fig. 1). The binary evolved has three asymptotic ends: two independent wormholes, each leading to its own far universe. A foam handle differs—its two mouths belong to one wormhole and carry opposite scalar charge by construction—so our opposite-signed pairs reproduce a handle's charges but not its topology. What merges is not two wormholes into one: both throats end behind one common horizon.

The observable is not a chirp but a population of bursts from the merger epoch, redshifted into whatever band the mass scale selects [25–28]. No throat survives more than milliseconds to days (Sec. IV G), so neither LIGO nor LISA [96–98] can see a wormhole in residence: a population born in the early universe converts at birth, its bursts redshifted far below both bands, and a drainhole signal in them would need throats born that late, which no known mechanism supplies. The epoch-independent deliverable is the energy radiated per event and its spec-

* Corresponding author: shirokov.nm@phystech.edu
† ommnammashivaya@gmail.com

trum in units of the source's ADM mass (Sec. IX B). Every waveform behind it is extracted in the near zone, on spheres at $R = 10$–44, and none is extrapolated to null infinity: each radiated energy carries the spread across its spheres as a near-zone error bar (Sec. VIII).

Sections II and III give the model and numerics; Secs. IV–VII follow the lone wormhole, two throats at rest, the head-on merger and the orbital channel; Sec. VIII treats the radiation of every channel and Sec. IX its observational signatures; Sec. X draws the consequences and Sec. XI gives reproducibility and cost. Appendix A holds the code-health record and Appendix B supplementary measurements. All evolutions use `GRTeclyn` [41], the AMReX-based [42] GPU port of `GRChombo` [43], continuing Refs. [44, 45]. We use $G = c = 1$.

II. THE MODEL

A. Field equations

The matter is a single real, massless phantom scalar: gravity couples to *minus* the canonical stress tensor,

$$T_{ab} = -\Big(\nabla_a\phi\,\nabla_b\phi - \tfrac{1}{2}g_{ab}\,\nabla_c\phi\nabla^c\phi\Big), \qquad V(\phi)\equiv 0, \quad (1)$$

so the energy density seen by normal observers, $\rho = -\frac{1}{2}\big(\Pi^2 + \gamma^{ij}\partial_i\phi\,\partial_j\phi\big)$ with $\Pi = (\partial_t\phi - \beta^i\partial_i\phi)/\alpha$, is non-positive everywhere, and the null energy condition fails wherever ϕ varies—the exotic stress that holds a throat open. The field itself obeys the canonical wave equation $\Box\phi = 0$, evolved in first-order form in ϕ and Π.

B. Drainhole initial data

Each body is a massive Ellis–Bronnikov drainhole [9, 10]. In isotropic radius r about the throat centre, with

$$X = \frac{r - a^2/4r}{a}, \quad \Omega = 1+\frac{a^2}{4r^2}, \quad u = \frac{m}{a}\Big(\arctan X - \frac{\pi}{2}\Big), \quad (2)$$

the exact static solution of Eq. (1) is

$$\alpha = e^u, \quad \gamma_{ij} = e^{-2u}\,\Omega^2\,\delta_{ij}, \quad \phi = C\arctan X, \quad (3)$$

with $4\pi a^2C^2 = a^2 + m^2$, $K_{ij} = 0$, $\Pi = 0$ and zero shift; in the evolved conformal variables $\chi = e^{2u}\Omega^{-2}$ and $h_{ij} = \delta_{ij}$. Both constraints hold identically at $t = 0$. Here a is the throat scale and m the ADM mass, carried by the lapse rather than the conformal factor, so χ at the throat stays in $[0.15, 0.25]$ for $m/a \le 1$ and mass costs no resolution. The minimal surface sits at $r_t = \frac{1}{2}\big(m+\sqrt{m^2+a^2}\big)$, where $X = m/a$, so its areal radius is analytic: $R_{\min} = e^{-u(r_t)}\sqrt{a^2+m^2} = \sqrt{a^2+m^2}\,\exp\big[(m/a)\arctan(a/m)\big]$, which is a at $m = 0$. The production throat $a = 2$, $m = 1$ has $R_{\min} = \sqrt{5}\,e^{\frac{1}{2}\arctan 2} = 3.8895$ at $r_t = \frac{1}{2}(1+\sqrt{5}) = 1.618$. The point $r \to 0$ is the far side's spatial infinity, compactified ($\chi \sim r^4$). At $m = 0$ the throat is ultrastatic ($\alpha \equiv 1$, $M_{\rm ADM} = 0$).

C. Two throats in one slice

Two throats at separation d (one-body radii r_A, r_B) start from their combined fields,

$$u = u_A + u_B, \qquad \psi = 1 + \big(\sqrt{\Omega_A} - 1\big) + \big(\sqrt{\Omega_B} - 1\big), \quad (4)$$

with $\chi = e^{2u}\psi^{-4}$, $\alpha = e^u$ and $\phi = \phi_A + \sigma\,\phi_B$, $\sigma = \pm 1$; the constant tail of the arctan profiles is subtracted so that $\phi \to 0$ at the boundary. This is the starting point of the constraint solve of Sec. II D, which turns it into initial data. Momentum is carried by the exact boost of Sec. II E: the throats sit at $x = \mp d/2$ with $P^i = (0, \pm p, 0)$, momenta of magnitude p in units of the one-body mass, tangential to the separation and opposite on the two throats, so the pair carries no net momentum and orbital angular momentum $J = pd$; $p = 0$ is a release from rest. The relative sign σ is the pair's one discrete parameter. Far from its throat each body is a mass m carrying the scalar charge $q = Ca$ ($\phi_A \simeq -q/r_A$ once the constant tail is subtracted), and a phantom mediator *repels* like charges and *attracts* opposite ones, so two throats at rest a large distance d apart attract with

$$F = \frac{m^2}{d^2}\,(1 - \sigma Q), \qquad Q \equiv \Big|\frac{F_\phi}{F_{\rm grav}}\Big| = \frac{4\pi q^2}{m^2} = \frac{a^2+m^2}{m^2} > 1 \quad (5)$$

always ($Q = 5$ for the production throat): a like-signed pair repels and never merges under its own gravity, and $\sigma = -1$ is the merger channel.

D. Constraint-solved data and matched placement

With $\Pi = 0$, $K = 0$ and $\gamma_{ij} = \Psi^4\delta_{ij}$ the Hamiltonian constraint of Eq. (1) reads, in flat operators,

$$\nabla^2\Psi - \pi|\nabla\phi|^2\,\Psi + \tfrac{1}{8}\hat{A}_{ij}\hat{A}^{ij}\,\Psi^{-7} = 0. \quad (6)$$

The single drainhole solves it exactly. For a pair we solve it for a correction w to the starting Ψ on the whole initial mesh hierarchy with AMReX's multigrid [42], holding ϕ, Π and the lapse fixed (a Robin outer boundary); a pair with momentum solves both constraints together (Sec. II E).

Which throat is each mouth? Near its centre $\Psi = c/r + d_c + O(r)$, and the inversion $r' = c^2/r$ turns this into an asymptotically flat end—the mouth's own far universe—with ADM mass and scalar charge

$$M_{\rm far} = 2c\,d_c, \qquad Q_{\rm far} = \frac{4Cc^2}{a}, \quad (7)$$

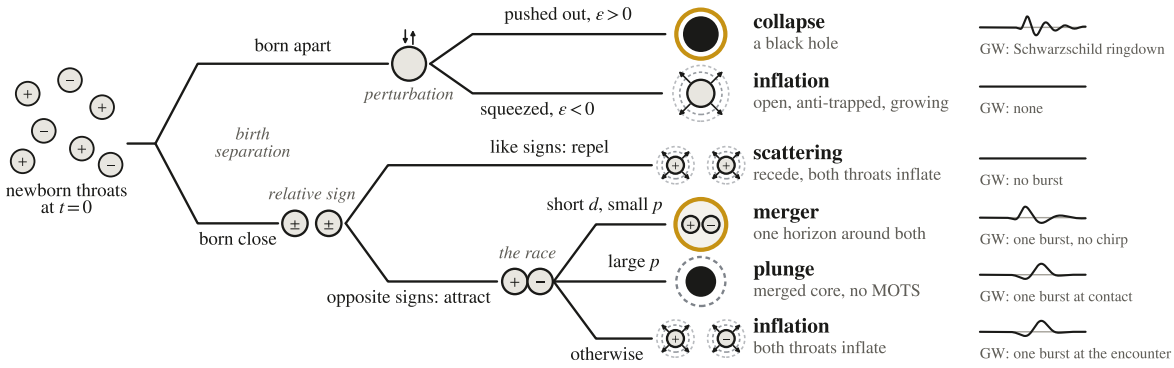


FIG. 1. Fates of newborn throats, whatever their origin, schematic: a lone throat's perturbation sign selects collapse or inflation (Sec. IV C), a pair's relative sign repulsion or attraction (Sec. V A), a repelling pair's throats inflate as it recedes (Sec. V C), and an attracting pair merges where its horizon outruns inflation (Sec. VII B) or, plunging fast, collapses to a merged core on which no common MOTS converges (Sec. VII A). Gold: horizons; right: sketched wave signals. The outcome: two fates end in a black hole, each after a single chirpless burst, and the fast plunge in a merged core; in every other fate the throats inflate.

TABLE I. Each mouth of the matched $d = 8$ pair at $t = 0$ ($L = 64$, level 3) against the isolated throat: minimal areal radius, far-side mass [Eq. (7)], one-body mass m', and the $M_{\rm ADM}$ of Eq. (8)—the throat's own, and the pair's (2.3573 on the production $L = 128$ grid).

data	$R_{\rm min}$	$M_{\rm far}$	m'	$M_{\rm ADM}$
isolated throat	3.890	−4.810	1	1.001
matched pair, each mouth	3.892	−4.810	1	2.362

both frozen in time, like a puncture's individual mass. The isolated drainhole has $M_{\rm far} = -m\,e^{\pi m/a} = -4.8105$ and $Q_{\rm far} = \sqrt{a^2+m^2}\,e^{\pi m/a}/\sqrt{4\pi} = 3.0344$. A pair $(M_{\rm far}, Q_{\rm far})$ names one isolated drainhole, and its near-side mass m' is the mouth's one-body mass.

A second throat in the slice rescales each mouth: near A the companion's constants multiply the throat's own field, $\Psi \to e^{-u_B(x_A)/2}\,\Psi$, and because ϕ fixes a static throat's coordinate size, the puncture coefficient c alone cannot undo this. The *matched placement* gives each throat a coordinate scale λ ($a \to \lambda a$ and $m \to \lambda m$ in ϕ, u and Ω; C is scale-free), sets $\lambda = (c/c_{\rm iso})^2$ with $c_{\rm iso} = \frac{a}{2}e^{\pi m/2a}$, which holds $Q_{\rm far}$ at its isolated value, and iterates c until $M_{\rm far}$ is isolated too. Each mouth is then locally the isolated static drainhole (Table I): its $R_{\rm min}$ lands within 0.06% of $R_\star$ without being imposed, and $m' = 1$. With momentum the iteration converges the same way ($d = 12$, $p = 0.12$: $R_{\rm min}$ within 0.04% of the lone boosted throat's).

The pair's mass. With $\Psi \to 1 + M_{\rm ADM}/2r$, Gauss's theorem on Eq. (6), whose punctures contribute $-4\pi c_X \delta_X$, gives

$$M_{\rm ADM} = 2\sum_X c_X - \frac{1}{2\pi}\int\Big(\pi|\nabla\phi|^2\Psi - \tfrac{1}{8}\hat{A}_{ij}\hat{A}^{ij}\Psi^{-7}\Big)d^3x, \tag{8}$$

which returns the exact throat's $M = 1$ to 0.14% and, unlike a fit to the monopole tail, does not read the outer boundary's offset of w as mass. For matched pairs at rest (Table II) the excess over two isolated throats follows

TABLE II. Mass of matched pairs at rest [Eq. (8)] against two isolated throats on the same grid ($L = 128$, $\Delta x = 0.0625$ at the throats), and Eq. (9). λ: each throat's coordinate scale.

	$\sigma=-1$ (attract)			$\sigma=+1$ (repel)		
d	λ	E_b	Eq. (9)	λ	E_b	Eq. (9)
8	0.857	+0.357	+0.367	0.906	−0.619	−0.616
12	0.906	+0.269	+0.274	0.931	−0.437	−0.433
16	0.931	+0.214	+0.217	0.946	−0.338	−0.335
24	0.955	+0.151	+0.152	0.962	−0.233	−0.231
48	0.979	+0.080	+0.080	0.980	−0.121	−0.120

$$E_b \equiv M_{\rm ADM} - 2m = -\lambda^2\,\frac{m^2}{d}\,\big(1+\sigma Q\big) \tag{9}$$

for both signs, to 3% at $d = 8$ and 0.5% at $d = 48$. It is the Newtonian binding $-\lambda^2 m^2/d$ plus the phantom cross energy $-\sigma\int\nabla\phi_A\cdot\nabla\phi_B\,d^3x = -\sigma\lambda^2(a^2+m^2)/d$ of two charges λq, with $\lambda \to 1$ as $d \to \infty$; the solve adds nothing else. In the merger channel $\sigma = -1$ the scalar term is positive and wins: the attracting pair weighs *more* than its parts, $E_b\,d \to +4$, and the repelling pair less, $E_b\,d \to -6$. Equation (9) is the ADM mass of data at fixed far-side constants, not the two-body potential; read as one, it gives the wrong sign in both channels. Equation (5) is the standard force for a ghost mediator, which reverses the sign of ordinary scalar exchange, and the evolution confirms it (1.462 ± 0.022 against $3/2$, Sec. V A). How the flux through each throat responds to the companion is left to a direct measurement.

Units of this paper. Masses and times are in units of the drainhole parameter $m = 1$. Every production binary evolves matched, solved data: each mouth's far side is the

isolated throat's, $m' = 1$ to 6×10^{-6} across the production set, so a quoted momentum p is per true one-body mass, with boost speed $v = p/\sqrt{1+p^2}$, and the pair's mass is the solve's $M_{\rm ADM}$ [Eq. (8) at rest, Sec. II E in motion; 2.3573 for the $d = 8$ head-on pair].

E. Throats with momentum

A throat with ADM momentum p is the exact drainhole boosted to $v = p/\sqrt{m^2+p^2}$ and cut at $t = 0$: the metric, K_{ij}, ϕ and Π together, with the boosted lapse and shift, so that the slice is an exact solution of both constraints and the scalar moves with its throat ($\Pi \neq 0$). On the slice every χ contour, the throat's included, is an ellipsoid contracted along the motion by $1/\gamma = m/\sqrt{m^2+p^2}$, as a boosted black hole's horizon is; the measured axis ratios follow this law to 6×10^{-5} at every p up to 0.45 (Fig. 2).

A pair is built from two boosted throats as in Sec. II D, with the companion's K_{ij} and Π windowed out inside each throat, and the solve then corrects both constraints, for w and a vector potential W ($\hat{A} \to \hat{A} + \mathbb{L}W$). The window acts only inside each throat, about the far side's compactified infinity, and the solve then restores both constraints on the whole mesh, so no unsolved defect is left to radiate: the first light-crossing carries at most 2.7% of any quoted energy (Sec. VIII). Three checks anchor it: on a lone boosted throat on the production grid the solve is idle (puncture correction $|w_0| = 4.5\times10^{-6}$, the far side isolated to 4×10^{-6}); on the rest pair it reproduces the scalar solve's c, σ and w_0 to 10^{-5}; and on the opposite-signed $d = 12$, $p = \pm0.25$ pair on the production grid the far sides come out the isolated throat's to 10^{-5}, each mouth's $R_{\rm min}$ lands 0.09% from the isolated value, and the solved throat-shell Hamiltonian is at most 7.8×10^{-5} rms across the refinement levels. The pair's mass is the boosted background's own ADM energy—each throat's $\gamma\lambda_X m$ and its puncture shift, by the full surface integral of the closed forms—plus the monopole of w on the box faces, since Eq. (8) needs conformally flat data with $K = \Pi = 0$; where both apply, the face value reads 2.8% below it. For the $d = 12$ pairs at $p = 0.25$–0.60 a full ADM surface integral of the $t = 0$ slice, which recovers a lone boosted throat's γm, reads 2.28–2.32, nearly flat in p and 2.4–3.2% above this prescription; their energies are quoted per that measured mass.

F. Declared seeds

The instability is probed with declared perturbations of the conformal factor only: a Gaussian shell on the minimal surface,

$$\psi \to \psi\big[1 + \varepsilon\, g(r)\big], \qquad g = e^{-(r-r_t)^2/w^2}, \quad w = a/4, \tag{10}$$

which changes the throat's areal radius by a fraction $\approx 2\varepsilon$, and a quadrupolar factor

$$\psi \to \psi\big[1 + \varepsilon_2\, g(r)\, P_2(\cos\theta)\big] \tag{11}$$

about the z axis through the throat. Neither seed is constraint-solved. K_{ij} and Π stay untouched, so the momentum constraint holds at $t = 0$; but the drainhole solves the linear equation $\nabla^2\Psi = \pi|\nabla\phi|^2\Psi$ ($\Psi^4 = \chi^{-1}$, flat operators), so the spherical seed leaves exactly $\mathcal{H} = -8\varepsilon\,\Psi^{-4}(1+\varepsilon g)^{-5}(\nabla^2 g + 2\nabla\ln\Psi\cdot\nabla g)$—confined to the shell and, the shell being narrow, not small there: its peak is 0.93–$0.94 \times 16\pi|\rho|$ at $\varepsilon = \pm10^{-2}$. The quadrupolar seed's defect is pure $\ell = 2$ at first order. CCZ4 feeds the defect into Θ and, within the first steps, into $\mathcal{M}$ (Fig. 11(b)). The defect does not pick the branch: carried through the constraint solve (Sec. II D) as the puncture coefficient $c = 1.1\, c_{\rm iso}$—the solve, unique at fixed ϕ, would erase a conformal seed—the $\varepsilon = +0.1$ kick is trapped already on its $t = 0$ slice (Sec. VII B), as the unsolved $+0.1$ seed is from $t = 1$ (Sec. IV C). Above $\varepsilon \simeq 0.025$ the spherical seed turns the minimal surface into a maximum between two minima. The spherical seed selects the branch but cannot radiate: a spherical collapse has no $\ell \geq 2$ moment. The quadrupolar seed, with no $\ell = 0$ part at first order, leaves the branch to it and gives a collapsing throat a quadrupole to shed—a lone throat's only gravitational-wave signal (Sec. VIII A).

III. NUMERICAL SETUP

A. Evolution scheme

We evolve CCZ4 [37] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, 1+log slicing [38] and a Gamma-driver shift [39, 40, 66] with $\eta = 1$, both started from the initial data's lapse and shift, fourth-order stencils, RK4 and Kreiss–Oliger dissipation $\sigma = 0.1$. The time step is $\Delta t = 0.02\,\Delta x$, an order of magnitude below the usual CCZ4 Courant factor, because larger steps fail: a multiplier of 0.1 dies at $t = 16$. Every resolution statement below is at fixed $\Delta t/\Delta x$; Appendix A tests convergence in resolution and in extraction radius.

B. Grid, boundaries and regularisation

The domain is a cube of side $L = 64$ centred on the system, with a 128^3 base grid ($\Delta x = 0.5$). Nested boxes, halving per level and centred on the tracked throats (half-width 4 per throat at the production finest level), are re-drawn every 16 steps. Production runs use three refinement levels (finest $\Delta x = 1/16$); resolution arms go to levels 4–7 where stated, and every binary deeper than level 3 and two lone-throat arms use $L = 128$: the same cells in a box twice as wide. Level ℓ thus means a finest cell $\Delta x = 0.5/2^\ell$. The one exception is the octant of an

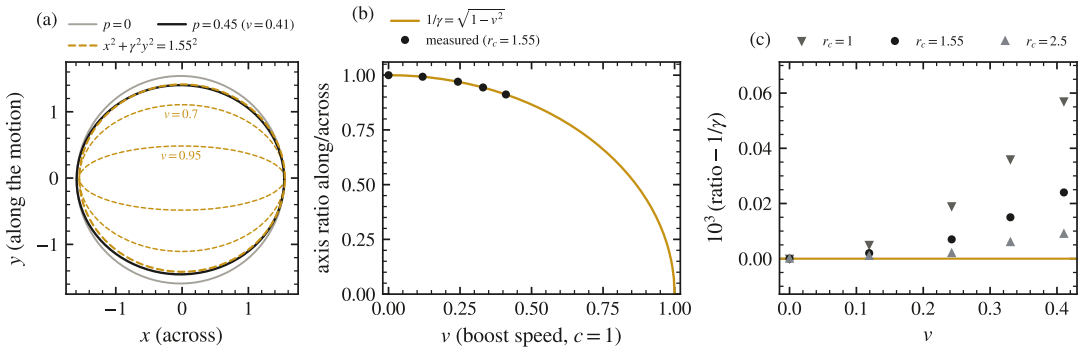


FIG. 2. The moving throat at $t = 0$ is the Lorentz-contracted drainhole. (a) The χ contour just inside the throat, at rest and at $p = 0.45$, the latter on its analytic ellipse; the fainter ellipses continue the law to $v = 0.7$ and 0.95. (b) Axis ratio (along/across the motion) against the boost speed $v = p/\sqrt{m^2 + p^2}$, on $1/\gamma = \sqrt{1 - v^2}$ drawn to the disc limit $v = 1$. (c) The residual: below 6×10^{-5} at every v and at three contour levels. The outcome: the boosted data carry exactly the special-relativistic shape, with the scalar in motion, and both constraints hold at the exact solution's discretisation floor.

TABLE III. The run matrix: 103 of the archived evolutions—the wormhole pairs on constraint-solved data, momentum carried by the exact Lorentz boost (Sec. II E)—in groups whose runs (arms) each vary one setting against a named reference or repeat one; per-run lines are in the shipped registry. Left: the single throat and the initial-data machinery. Right: the binaries and their controls. Unless stated, $a = 2$, $m = 1$ ($R_{\min} = 3.89$), $\sigma = -1$, $d = 12$ and $L = 64$ (128 for binaries deeper than level 3).

group	knob(s)	runs	Sec.	group	knob(s)	runs	Sec.
lone throat	levels 2–4; χ floor; Δt; L=128; m=0.5	10	IV	two throats at rest	σ; a = 1–3; d = 12–18; one like pair to t=100	9	V
spherical kicks	$\varepsilon = \pm 10^{-3} \ldots \pm 10^{-1}$; levels 3–5	10	IV C	head-on, d=8	segments, levels 4–6; MOTS replay; first-law window	8	VI
quadrupolar kicks	$\varepsilon_2 = 0.005$–0.05; kick-free arm	6	VIII A	merger chain, d=6	p=0.10 tangential; levels 3–7; $\sigma_{\rm KO}$; χ floor; final segment	7	VII C
scalar-stream re-runs	unkicked level 4, L=64/128; ε=+10^{-2}	3	VIII G	d=12 arms	p = 0.12 tangential, 0.25 inward	2	VII B
boosted throat	p = 0–0.45 at t=0; collapse twins, levels 3/4	7	II E	fly-by and $E_{\rm GW}$ scan	p = 0.25–0.90; χ-floor and level-5 extensions	8	VIII D
solve verifications	boosted pairs at t=0 on the production grid; the pair at level 3 to t=20; a kicked single through the solve	4	II E	vacuum BBH	same d, p, masses; no scalar, no sponge	5	VIII E
placement	d = 6–48, one step each	18	II D	convergence	head-on at three resolutions, L=512; $\sigma_{\rm KO}$; d=12 arm at level 4, wave zone refined	6	A

$L = 512$ box that carries the inflating throat (Sec. IV F): its base spacing is 2, so its 5 levels give $\Delta x = 1/16$, the resolution of our level-3 arms, and we quote it by that Δx. Outer boundaries are Sommerfeld, backed by a spherical sponge about the box centre over the outer quarter of the half-width—$r = 24$–32 at $L = 64$ and 48–64 at $L = 128$—in which extra dissipation ramps to $\sigma_{\rm sp} = 4$ as the fourth power of the depth; every extraction sphere lies inside its inner radius. The sponge is Kreiss–Oliger dissipation only: it damps grid-scale noise but is transparent to long-wavelength gauge and physical waves, which reflect from the box faces (Sec. VI B). The vacuum controls (Sec. VIII E) instead run `GRTeclyn`'s binary-black-hole example with its own numerical settings and no sponge.

Each throat carries a compactified second infinity at its centre, where $\chi \sim r^4$, so the evolved variables are floored at $\chi \geq 10^{-8}$ and $\alpha \geq 10^{-10}$ (a variant floors χ at 10^{-20} and applies 10^{-8} only where $1/\chi$ is evaluated); χ-floor twins at 10^{-8} and 10^{-20} re-converge to four digits after touching the floor. Where Table IV says so, the 1+log source is off within coordinate radius 0.3 of each tracked throat centre, the far side's compactified infinity, and restored by 0.8 ($r_t = 1.618$), so that the moving puncture does not drive the slicing, and the scalar is damped exponentially wherever the lapse has collapsed below 3×10^{-2}: inside merged cores, and between fly-by mouths as they approach. The $p = 0.45$ fly-by rerun undamped to $t = 70$ matches its damped twin's Ψ_4 to 2×10^{-7} of the peak. The damping first acts at $t = 36$, so this comparison covers only $R \lesssim 34$: the burst peak at $R = 20$, not the

TABLE IV. Numerical modifications per run beyond Sec. III A ($\sigma_{\rm KO} = 0.1$, $\chi \geq 10^{-8}$, 1+log everywhere, no damping). Restart segments follow semicolons, their start times in parentheses; V is the variant χ floor; last column: 1+log off near the throats (O), scalar damped (D).

Run	Levels	χ floor; $\sigma_{\rm KO}$	O, D
Lone throat	2–4	twin V	–
Moving throat	3, 4	–	O
Rest pairs	3	–	–
Head-on	5; 6 (35); 4 (50)	V; twin σ 0.3	–
Far zone	5, 6, 7	10^{-4}; σ 0.3	–
$d = 6$ chain	5; 5 (25, 30); 4 (35)	σ 1.0 (25); 10^{-4} (30)	O, D
$p = 0.12$	4, 5	–	–
$p = 0.25$	5	–	O, D
$p = 0.45$	4	–	O, D
$p = 0.60$	4; 4 (40, 50); 5 (40, 60)	10^{-4} (40)	O, D
$p = 0.90$	4; 4, 5 (40)	10^{-4} (40)	O, D

$R = 28$ window the energy is read from.

C. Diagnostics

Every coarse step the code writes the volume-averaged norms $\|\mathcal{H}\| = (\int \mathcal{H}^2 \,\mathrm{d}V/L^3)^{1/2}$ and $\|\mathcal{M}\| = (\int \mathcal{M}_i\mathcal{M}_i \,\mathrm{d}V/L^3)^{1/2}$, evaluated with fourth-order differences on the base grid over the whole domain after the finer levels are restricted onto it: they do not resolve the throats, monitor code health rather than local accuracy, and compare only within one box. At $t = 0$ they are set by the base grid's stencils on the throat profiles, not by the data: the exact throat reads 2.1×10^{-3} at levels 2–4 alike, and the $d = 12$ pair 1.1×10^{-3} at levels 4 and 5. Both throat centres are tracked, and about each the areal radius $R(r)$ is scanned on coordinate spheres out to a fixed edge; its minimum is the throat, and an inflating throat that outgrows the scan is bounded from above by the edge reading. On the same spheres the null expansions $\theta_\pm$ are computed with the outward direction fixed by *increasing areal radius*, so that a dynamic throat is marginally *anti*-trapped [35, 64]; the in-code monitor orients by coordinate radius, which points the wrong way beyond a throat, and serves only as a live monitor.

The star-shaped scan reports the outermost coordinate sphere that is trapped everywhere ($\max\theta_+ = 0$, $\theta_- < 0$). A trapped sphere lies inside the apparent horizon, so this sphere—"the scan's MOTS" below—bounds the marginally outer trapped surface from inside. On its spheres, as on the finder's surfaces of Sec. III D, we evaluate the areal radius R and the Hawking mass [34], the Misner–Sharp mass [33] on round spheres,

$$\frac{2M_{\rm MS}}{R} = 1 + \frac{R^2\,\langle\theta_+\theta_-\rangle}{4}, \tag{12}$$

with the product averaged over the sphere and $\theta_\pm$ computed for the null normals $k^a = n^a + s^a$ and $l^a = n^a - s^a$ (n^a the slice's unit normal, s^a the sphere's outward unit normal, so $k \cdot l = -2$); with the normalisation $k \cdot l = -1$ the factor $1/4$ becomes $1/2$. On an exact MOTS $M_{\rm MS} = R/2$; on the scan's spheres $M_{\rm MS}$ exceeds $R/2$ by at most 3% about a lone collapsed throat, which stays nearly round.

Radiation is read from Ψ_4 on spheres at $R = 10$, 14, 18 (and 22 on the single-throat wave arms) on the single throats, and at $R = 20$, 28, 36, 44 on every $L = 128$ binary, the head-on adding the single throats' three, decomposed in spin-weight -2 harmonics about z, the orbital axis. The head-ons fall along x: there the z-based $(2, 0)$ is $-h'_{20}/2$ of the quadrupole h'_{20} about the collision axis and carries a quarter of its power, the $(2, \pm2)$ the rest ($|h_{22}/h_{20}| = \sqrt{3/2}$ to 2% on every sphere of both head-ons), so their $(2, 0)$ is quoted as h'_{20}. Every amplitude quoted is $r\Psi_4$, and strain is its fixed-frequency integral [93]; on the head-on the in-code and offline pipelines agree at the burst peak to 0.4%, while on the single throats the in-code stream is dominated by its grid-scale floor and only the offline one is used. A scalar stream is read on the single throats' spheres and, on the binaries, at $R = 14$, 20, 30 and 44 (the head-on adding 10 and 18), where the offline pipeline also reads Ψ_4: ϕ and Π in spin-weight-0 harmonics, $A_{\ell m} = R\,\phi_{\ell m}$, and the kinematic flux $F_{\rm kin} = -R^2 \oint \Pi\,\partial_r\phi \,\mathrm{d}\Omega$, positive for an outgoing *canonical* wave (Sec. VIII F). An opt-in per-cell autopsy records the location and one-step history of any floating-point failure.

D. Marginally outer trapped surfaces

A marginally outer trapped surface (MOTS) is a closed two-surface of the slice on which the outgoing null expansion vanishes, $\theta_+ = 0$, while the ingoing one is negative, $\theta_- < 0$; the outermost MOTS is the apparent horizon. A MOTS is read off a single slice, unlike the event horizon, and since the phantom violates the null energy condition it need not lie inside one: every "horizon" and "censored" in this paper is a statement about MOTSs on the evolved slices, and "black hole" names the region inside such a dynamical apparent horizon, whose area this matter can shrink (Sec. IV E), not a stationary black hole, which the remnants only approach (Sec. VI B).

The finder runs in post-processing on every archived slice, so horizon coverage matches the plot cadence. It expands the surface about the tracked centre as $r = h(\theta, \phi)$ in real spherical harmonics up to degree $\ell_{\rm max} = 6$ on the head-on slices and $\ell_{\rm max} = 8$ on the orbital ones, and solves $\theta_+[h] = 0$ by Newton iteration on metric and extrinsic-curvature data interpolated from level 3 on the head-on and level 2 on the orbits, whatever the run's finest level, with the areal orientation of Sec. III C. The first surface of a run is found by the damped fast flow [65]; each later slice warm-starts from the previous surface and converges in a few iterations until every projected harmonic of θ_+ is below the solver tolerance of 10^{-6}. From the converged surface we quote the areal radius

$R = \sqrt{A/4\pi}$, the Misner–Sharp mass of Eq. (12), and the harmonic content beyond $\ell = 0$ as the deformation. Its flow stage recovers Kerr–Schild Schwarzschild ($R = 2M$ to 0.24%, $M_{\rm MS} = M$ to 0.12%) and finds no trapped surface outside the time-symmetric Ellis throat. Every binary horizon number in this paper—formation times, R, $M_{\rm MS}$—is read from these surfaces, the lone throats' nearly round collapse horizons from the star-shaped scan (within 3%, Sec. III C). On a binary the scan is only an inner bound and a live monitor: a fully trapped round sphere lies strictly inside a deformed MOTS, so the scan under-reads a forming horizon, or misses it.

IV. THE SINGLE WORMHOLE

A. The static throat as a fixed point

Evolved without a declared perturbation, the exact drainhole holds $R_\star = 3.8895$ to 0.1% for 35 time units at level 3 and to 1% until $t \simeq 44$ (Fig. 3(a)); at level 4 it is exact to four digits at $t = 30$. The departure moves *later* by 5.6–8.7 units per halving of the cell—a truncation-noise seed of fixed growth rate handed a smaller amplitude. Level 2 dies of an origin failure at $t = 24.2$ with the throat still exact to 0.12%. The hold scales with the mass: at level 3 the $m = 0.5$ throat holds 0.1% only to $t \approx 17$, against 35 at $m = 1$, and dies at its origin at $t = 26.3$.

B. The unstable mode

The departure is a single exponential mode (Fig. 3(a,b)): fits below 4% deviation give $\tau = 5.12$ (level 3) and 5.23 (level 4), i.e. $T \equiv \tau\,\alpha_{\rm th}/R_\star = 0.76$ and 0.77, where $\alpha_{\rm th} = e^{u(r_t)} = 0.575$ is the static lapse at the throat, so that T is the proper-time e-fold in units of the throat's light-crossing time. Our drainhole is the $\gamma_1 = m/a = 0.5$ member of the static family of González, Guzmán and Sarbach (their metric with $b = a$, $\gamma_0 = -\pi\gamma_1/2$), for which their linear analysis gives $T = 0.758$; their Table I spans $T = 0.846$ at the massless end to 0.590 [18]. Solving the linearised field equations of this solution ourselves—their regularised master equation and, independently, the reduction in the gauge $\delta\phi = 0$—gives $T = 0.7584$, i.e. $\tau_{\rm lin} = 5.13\,M$. The measured e-fold matches it at both resolutions, to 0.2% at level 3 and 1.9% at level 4; larger departures grow more slowly. The seed is truncation error, and its sign flips with resolution—level 3 collapses, level 4 inflates—the two branches found in spherically symmetric evolutions of the massless [13, 14] and massive [15, 17] throat, and in 3D for the massless one [44], here reached from exact massive data in 3D.

C. Branching: collapse versus inflation

A declared kick, Eq. (10), overrides the truncation seed (Fig. 3(c)). At $\varepsilon = \pm 10^{-2}$ and $\pm 10^{-3}$ the fate is always opposite to the push, each $\pm$ pair crosses at the ε-independent $t_\times \simeq 13.0$, and a tenfold smaller kick arrives 11–12 units later at matched horizon radius—linear response on the same mode. The branching survives refinement: at level 4, whose own seed inflates the unkicked throat, $+10^{-2}$ still forms a horizon at $t = 11$ and -10^{-2} still inflates, the twins crossing at $t = 12.99$ against level 3's 13.01. The branch point is the throat's, not the grid's or the kick's. The kick also sets the clock: horizons form at $t = 11$ ($\varepsilon = +10^{-2}$) and $t = 25$ ($+10^{-3}$), against $t \le 61$ for the unkicked throat (its first scanned slice after departure already holds a MOTS). At $\varepsilon = \pm 0.1$ the data are no longer perturbative (Fig. 3(c)): $+0.1$ turns the minimal surface into a maximum between two minima and is trapped from $t = 1$, while -0.1, which narrows the throat to $R_{\rm min} = 3.15$, re-expands to 3.36 by $t = 4$ and then collapses as well, trapped from $t = 6$—against the sign rule. Both die at the compactified origin ($t = 14.1$ and 15.2, NaN in h_{11}), the failure that also ends the $+10^{-3}$ arm at $t = 40.1$, before the lapse has collapsed anywhere. Their Hamiltonian defect is 3–6× that of the 1% arms in the box norm and 8.6–10.3 × $16\pi|\rho|$ on the shell; the usable range ends at 10^{-2}.

D. The throat in motion

A uniformly moving throat keeps the resting throat's fate. The exact-boost $p = 0.45$ throat (Sec. II E; level 3, $v = 0.41$) rides $R_\star$, rising at most 1.2%, until $t = 36.8$, where it falls 1% below its start (the resting level-3 throat: $t \simeq 44$), and then collapses on the resting throat's own mode: $\tau = 5.0$–6.1 against 5.12 at rest, on the same branch—level 3 collapses—and it dies at the same numerical endpoint ($t = 44.67$, Fig. 3(d)). The instability is the throat's, carried with the motion: time dilation ($\gamma = 1.10$) predicts 5.6 in coordinate time, which the fitted 5.0–6.1 cannot separate from 5.12; the branch and, within the fit, the proper-time rate are unchanged.

E. The collapse endpoint

The collapse branch ends in a black hole that consumes its supporting field. On the unkicked level-3 arm the horizon's areal radius falls 3.23 → 2.37 and $M_{\rm MS}$ 1.62 → 1.23 over 39 units, with the origin lapse at 0.016; the kicked $\varepsilon = +10^{-2}$ arm runs clean to $t = 100$ behind its horizon. Unlike the massless throat of Ref. [44], which bounced at $t \approx 4M$ and shed its horizon by $18.5M$, no arm here sheds a horizon. That collapse was driven with the phantom's stress-energy halved in the Einstein equations; its bounce and horizon shedding do not occur in the unmodified evolutions here and are artifacts of the halved stress-energy.

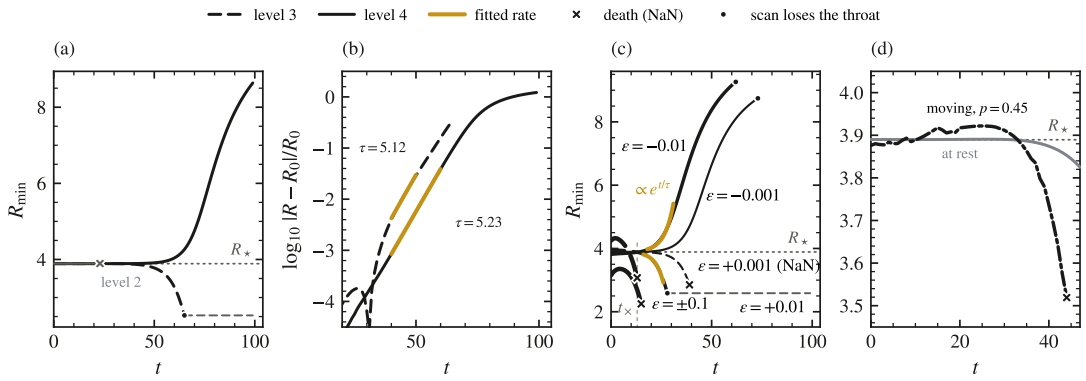


FIG. 3. Instability of the isolated throat. (a) Minimal areal radius from exact static data at refinement levels 2–4. The exact solution ($R_\star$, dotted) is held until truncation noise excites the unstable mode: level 3 (dashed) collapses, level 4 (solid) inflates, level 2 (grey) dies of an origin failure first (cross). Dots mark where the scan loses the throat; muted lines carry the last reading. (b) $|R-R_0|/R_0$: one exponential mode with a level-dependent sign; gold, small-amplitude fits, $\tau = 5.12/5.23$ at levels 3/4, the linear rate; larger departures slow. (c) Level 3 with declared seeds, Eq. (10): $\varepsilon = \pm10^{-2}$ (heavy), $\pm10^{-3}$ (light) and, heaviest, ±0.1, which both collapse; dashed, outward kicks. Twins cross at $t_\times \simeq 13.0$ independent of ε. Gold: local exponential fits; crosses, at the last reading: deaths at the compactified origin, $t = 40.1$ ($+10^{-3}$), 14.1 ($+0.1$) and 15.2 (-0.1). Their constraint norms are Fig. 11(a,b). Unless a seed is declared, the grid's truncation noise picks the branch. (d) The throat in motion (Sec. IV D): the exact Lorentz-boosted throat (dash-dot, $p = 0.45$, level 3) against the resting level-3 arm (grey). It rides $R_\star$ to $t \approx 25$ and then collapses on the level-3 mode, dying at $t = 44.67$: the fate and its rate are the resting throat's, carried with the motion.

The pure-quadrupole arm ($\varepsilon_2 = 10^{-2}$, no radial kick) is fully instrumented (Fig. 4): a MOTS from $t = 33$ at $R = 3.80$, shrinking to 2.50 by $t = 60$ ($M_{\rm MS}$ 1.90 → 1.25), while $|K|$ grows inward and the lapse pinches to zero at the centre.

The shrinking is physical. In spherical symmetry a massless phantom can only shrink a horizon: with the double-null metric $-2e^{-f}\mathrm{d}u\,\mathrm{d}v + R^2\mathrm{d}\Omega^2$, $T_{uv} = 0$ and $T_{vv} = -(\partial_v\phi)^2$, the Misner–Sharp mass on a marginally trapped tube obeys $\partial_v m = 4\pi R^2 e^f(\partial_v\phi)^2\partial_u R \le 0$ and $\partial_u m = 0$, and the tube is timelike or null [56, 63], so $R = 2m$ can only fall, as phantom accretion requires [57, 74]. The arms kicked with $\varepsilon = +10^{-2}$ shrink by 40% in areal radius to a floor at $t \approx 47$; the regrowth their scans show after it tracks the unconstrained seed's Hamiltonian defect (Sec. II F) reaching the horizon, and is absent from the pure-quadrupole arm, whose seed has no $\ell = 0$ defect (Fig. 4(a)). The throat becomes a black hole that swallows its own phantom support and loses 40% of its horizon radius doing so, in minutes even at $10^6\,M_\odot$ (Table V): Hawking's area law assumes the null energy condition [36, 58], and this horizon runs it backwards.

F. The inflation branch

The inflation branch forms no trapped surface. The $\varepsilon = -10^{-2}$ throat, evolved on an octant of an $L = 512$ box with finest $\Delta x = 1/16$ (the resolution of our $L = 64$ level-3 arms; Fig. 4), keeps growing to the end of the record, ×3.8 by $t = 218$, and stays anti-trapped [64]: the $\theta_l = 0$ horizon sits on the neck and $\theta_k = 0$ runs outward, to $R \approx 73$ by $t = 212$. Until $t \approx 40$ the neck grows exponentially in its own proper time with $HR_0 = 1.22$ (locally 1.0–1.3; $R_0 = 3.81$, the kicked neck), 6% below our linear mode's 1.30, which its local rate reaches. Spherical evolutions find the same proper-time exponential: $Ha \approx 1.1$ for the massless throat [13] and exponential growth for massive ones [17]. After that the 1+log slicing freezes the lapse at the neck (0.57 → 0.015), only ≈ 5 units of its proper time pass by $t = 218$, and its growth per unit t slows—a property of the slicing, not of the throat.

The throat is shedding its own support: the monopole scalar flux through every sphere grows quasi-exponentially over the quotable window (Sec. VIII G), drawing on a store that outside the static throat is a Misner–Sharp deficit of $R_\star/2 - m = 0.94$; while the exterior slicing is still quiet the arms of Sec. VIII G shed 0.03 of it through $R = 18$, a few percent. By $t = 218$ neither a downturn of $R_{\min}(t)$ nor a trapped surface occurs: the growth holds at every sample, and the $\theta_k = 0$ surface keeps moving outward.

G. Lifetime and physical scale

At fixed m/a the solution is scale-free, and the measured e-fold is $\tau\alpha_{\rm th} = 0.76$–$0.77\,R_\star$ in proper time—one light-crossing of the throat. The lifetime is a logarithm of the seed: the matched-radius delay of Sec. IV C sets a decade's cost at 11–12 $M \simeq \tau \ln 10$ (the first MOTSs differ by $14\,M$, the weaker kick's born smaller), and the unkicked throat's places the level-3 truncation seed at

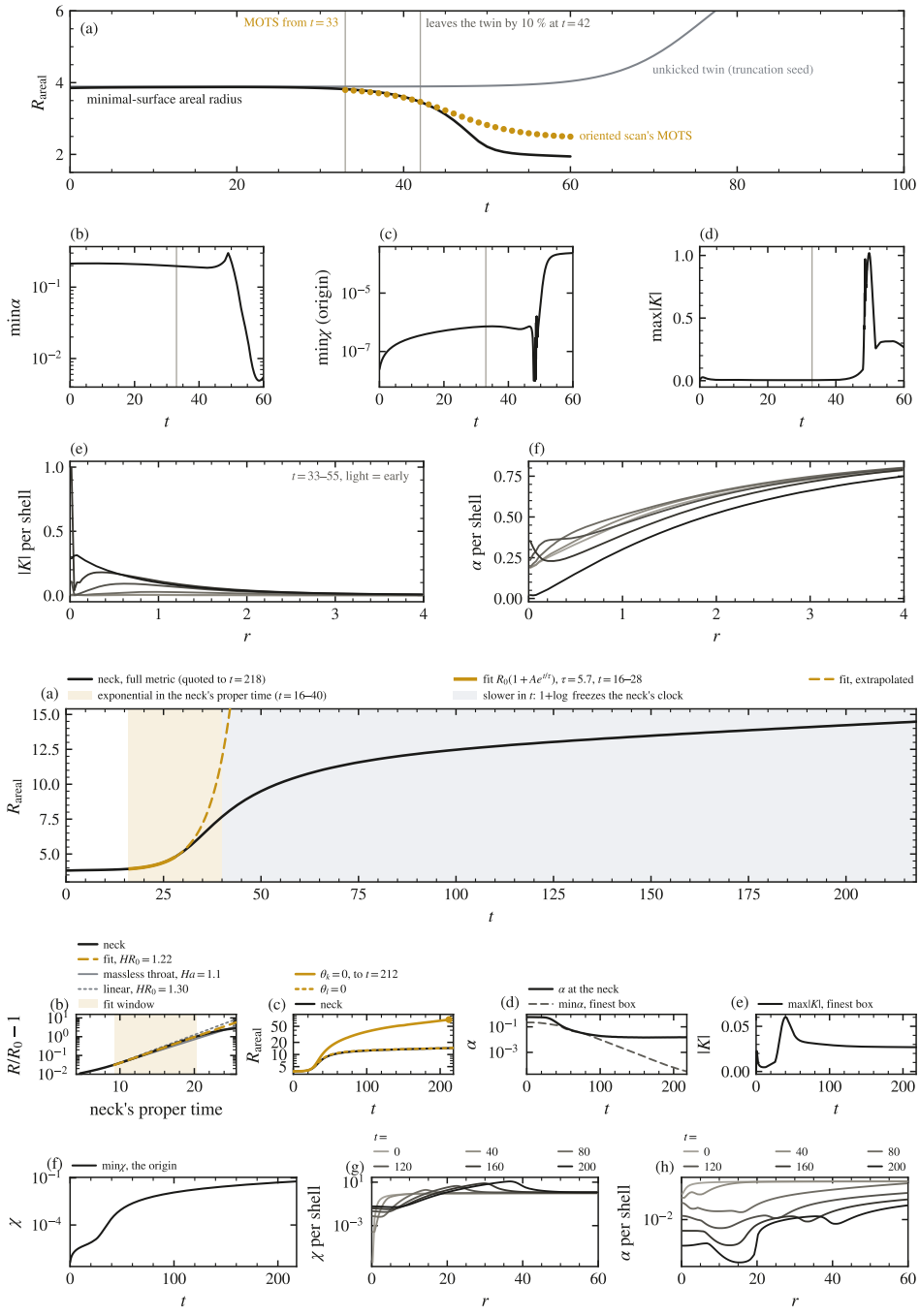


FIG. 4. *Top (collapse branch):* Collapse of the pure-quadrupole arm ($\varepsilon_2 = 0.01$, no radial kick). (a) Minimal-surface areal radius (ink) and the scan's MOTS (gold) from $t = 33$ to 60: R 3.80 → 2.50, $M_{\rm MS}$ 1.90 → 1.25. Grey: the unkicked level-4 twin, which inflates on its own seed; this arm leaves it by 10% only at $t = 42$—the second-order drive of the pure quadrupole. (b)–(d) Global $\min\alpha$, $\min\chi$ (the origin, the far side's compactified infinity) and $\max|K|$; the violent phase spans $t = 45$–55. (e),(f) Radial profiles over 128 shells, $t = 33$–55 (light = early): $|K|$ grows inward while α pinches to zero at the centre. Constraint norms: Fig. 11(c). The collapse ends in a black hole that consumes its supporting field. Movie: Video 1; youtu.be/Cc87mHcBPOA. *Bottom (inflation branch):* The inflating throat ($\varepsilon = -10^{-2}$; $L = 512$ octant, finest $\Delta x = 1/16$) to $t = 218$. (a) Neck areal radius in the full metric, 3.81 → 14.5; gold, the exponential onset, $\tau = 5.7$ over $t = 16$–28; shaded, exponential growth in the neck's proper time ($t = 16$–40, gold) and the slower growth in t once 1+ log freezes the neck's clock (grey). (b) Over that window the neck's proper-time rate is $HR_0 = 1.22$ (local 1.0–1.3), against the massless throat's $Ha = 1.1$ [13] and the linear mode's 1.30. (c) The $\theta_l = 0$ horizon stays on the neck; $\theta_k = 0$ runs out to $R \approx 73$ by $t = 212$. (d) The lapse at the neck falls 0.57 → 0.015; dashed, its minimum on the finest box. (e),(f) $\max|K|$ and $\min\chi$ on the finest box, around the origin. (g),(h) χ and α per shell (light = early): the χ peak moves out with the neck. Constraint norms: Fig. 11(d). The throat keeps growing, anti-trapped. Movie: Video 2; youtu.be/RIQCitu4tSc.

TABLE V. The drainhole clock in physical units: the production throat ($R_\star = 3.89\,M$) scaled by its ADM mass M. $\tau = 5.12\,M$ is the e-fold time (Sec. IV B); the last three columns are the measured times to a horizon for the two declared kicks, both outward ($\varepsilon > 0$, the collapsing sign; Fig. 3(c)), and an upper bound for truncation noise (Sec. IV C). Intermediate seeds follow $t_{\rm BH} \simeq 11\,M + \tau \ln(10^{-2}/\varepsilon)$ at matched horizon radius: $22.8\,M$ at 10^{-3}, whose first MOTS, born smaller, comes at $25\,M$.

			time to a horizon		
M	$R_\star$	τ	$\varepsilon = +10^{-2}$	$+10^{-3}$	noise ($\lesssim$)
$30\,M_\odot$	172 km	0.76 ms	1.6 ms	3.7 ms	9.0 ms
$10^3\,M_\odot$	5.7×10^3 km	25 ms	54 ms	0.12 s	0.30 s
$10^6\,M_\odot$	5.7×10^6 km	25 s	54 s	2.1 min	5.0 min
$10^9\,M_\odot$	38 AU	7.0 h	15 h	1.4 d	3.5 d

$\varepsilon \gtrsim 6 \times 10^{-7}$. Table V converts these to physical units. No static throat of astrophysical mass survives: a stellar-mass drainhole is gone in milliseconds, a supermassive one in minutes, and only at $10^9\,M_\odot$ does the lifetime reach days. The logarithm makes this robust: ten further decades of quiet initial data extend a $30\,M_\odot$ throat from 1.6 to 19 ms. A foam-born pair must collide as it is born; otherwise each throat collapses or inflates on its own first (Sec. X A).

H. Type-I criticality

In the language of critical collapse the static throat is consistent with a type-I critical solution [82]: a static solution with a single unstable mode (Sec. IV B), sitting on the threshold between two end states—collapse and inflation, where the standard problem has collapse and dispersal. The measurements above are consistent with both type-I signatures, though two amplitudes do not establish the scaling. The first is time scaling: data a distance ε from threshold stay near the critical solution for a time $\tau \ln(1/\varepsilon)$ set by its one e-fold, which is the lifetime law of Table V, $t_{\rm BH} \simeq 11\,M + \tau \ln(10^{-2}/\varepsilon)$, and both branches leave along the same mode, each $\pm\varepsilon$ pair crossing at the same $t_\times$ (Sec. IV C). The second is a mass gap: black-hole formation turns on at finite mass, as at the static Bartnik–McKinnon solution of Yang–Mills collapse [83], not at the vanishing mass of type-II scaling. The $\varepsilon = +10^{-2}$ and $+10^{-3}$ horizons form at $R = 3.88$ and 3.81, both close to the throat's own $R_\star$: a tenfold smaller kick leaves the black hole's mass at formation almost unchanged, where type-II scaling would shrink it as a power of ε. Shinkai and Hayward found evidence, in spherical evolutions of the massless throat, for critical solutions with a definite black-hole mass [13]; here the critical solution is the static throat itself, its one unstable mode measured in 3+1 dimensions.

V. TWO THROATS AT REST

A. The sign rule

Two identical throats released from rest push apart; reversing one throat's scalar, $\sigma = +1 \to -1$ in Eq. (5), turns the push into a pull (Fig. 5(a,b)). Both forces share one radial law, so their ratio Q is separation-independent: no gap, mass or approach speed lets a like-signed pair fall together, and $\sigma = -1$ is the only gravity-driven merger channel. The flip is not a tuning: $\phi \to -\phi$ leaves T_{ab}, and with it the one-body geometry, unchanged. The measured pull-to-push ratio at $a = 2$ is 1.462 ± 0.022 (slice-to-slice scatter over 15 slices only) against the point-charge, large-separation $(Q+1)/(Q-1) = 3/2$ of Eq. (5). The throat mass is equal in all arms, but λ (0.906 against 0.931) and $M_{\rm ADM}$ (2.27 against 1.56; Table II) differ between them, a systematic outside that error.

B. The force law

How the push falls with distance is read off a ladder (Fig. 5(c)). Four like-signed pairs, identical but for the gap $d = 12/14/16/18$, are released from rest, and by a common $t = 11.5$ their separations have grown by $\delta d = 0.4791/0.3716/0.2963/0.2406$. For displacements this small, δd at a fixed time is proportional to the force, so under the inverse-square law of Eq. (5) $\delta d\, d^2$ would be the same on every rung. It is not: it rises monotonically, $69.0 \to 78.0$, across the ladder, which excludes inverse-square in the coordinate separation. Instead $F \propto (d+\delta)^{-2}$ with $\delta \approx 2.2$–2.8 describes it: δ fitted on $d = 12$–16 alone predicts 0.242 at $d = 18$, where inverse-square gives 0.213 and the measurement is 0.2406; the full-ladder fit gives $\delta = 2.65$. Since δ is a sizable fraction of the mouth radius $R_\star = 3.89$, the force is inverse-square in an *effective* separation exceeding the coordinate one by most of a throat radius. Across widths, at $d = 12$, the push grows at every rung (Fig. 5(d)): $\delta d = 0.1623/0.3250/0.4791/0.7583$ by $t = 11.5$ for $a = 1/1.5/2/3$, a power law $\delta d \propto a^n$ with $n = 1.2$–1.4 when fitted at $t = 8$–11.5, where point charges, whose net push $(Q-1)\,m^2/d^2$ is a^2/d^2 by Eq. (5), give $n = 2$.

C. The repulsion branch

A like-signed pair keeps receding, and its throats end as a lone throat squeezed inward does: both inflate (Video 12; youtu.be/FnbDs-0n5zw). Released from rest at $d = 12$, the pair's separation grows from 11.9 to 26.4 by $t = 60$, and each mouth, the mirror of the other, opens from $R_{\rm min} = 3.88$ to 4.78 by $t = 30$ with both null expansions positive on its minimal surface: anti-trapped, on the inflation branch of Sec. IV F. The finder holds no

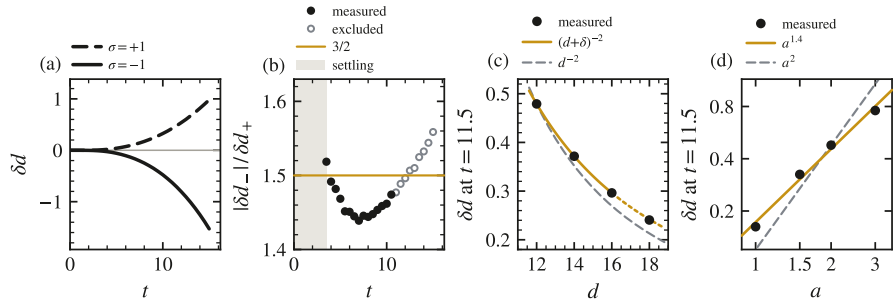


FIG. 5. Two throats at rest, on the far-side-matched pairs (Sec. II D): the sign rule (a,b), the force law (c) and its growth with throat width (d). (a) Separation change of two matched $d = 12$ pairs one sign apart: like-signed ($\sigma = +1$, dashed) opens, opposite-signed ($\sigma = -1$, solid) closes. Separations are inverse-χ-weighted centroids of the midplane χ pits. (b) Pull-to-push ratio against the prediction $(Q+1)/(Q-1) = 3/2$ (gold), where $Q = |F_\phi/F_{\rm grav}| = (a^2 + m^2)/m^2 = 5$ is the scalar-to-gravity force ratio of Eq. (5): measured 1.462 ± 0.022 (scatter over 15 slices); grey, the gauge-settling window $t < 3.5$, excluded; open circles past $t = 10.5$, where the closing pair has closed 4.6% of its gap, excluded too. (c) The force law on the ladder: how far like-signed pairs released from rest at $d = 12/14/16/18$ have pushed themselves apart by $t = 11.5$ (dots), against inverse-square in the coordinate separation scaled through $d = 12$ (grey, dashed) and $A/(d+\delta)^2$ fitted on $d = 12$–16 alone (gold), dotted where it predicts the $d = 18$ rung (Sec. V B). If the push were inverse-square in d, every dot would lie on the grey curve; the dots rise above it, and the gold fit absorbs the excess with the offset δ. (d) The push against throat width, on logarithmic axes: how far like-signed $d = 12$ pairs of width $a = 1/1.5/2/3$ have pushed themselves apart by $t = 11.5$ (dots; $a = 2$ is the $d = 12$ rung of (c)), against the point-charge a^2 of Eq. (5) scaled through $a = 2$ (grey, dashed) and a power law fitted to all four rungs (gold). If the push grew as a^2, every dot would lie on the grey line; the dots climb more slowly, as $a^{1.4}$.

MOTS about the mouth after $t = 16$, and the pair radiates no burst: its $\ell = 2$ Ψ_4 keeps one sign and drifts smoothly through the trust window. The companion's squeeze that starts every pair's inflation (Sec. VII B) acts here with no contact to race against.

VI. THE HEAD-ON MERGER

A. Contact and the common horizon

Released from rest—$p = 0$—at $d = 8$ with opposite scalar signs, on matched, solved data ($M_{\rm ADM} = 2.3573$), the pair falls together, and the spectral finder (Sec. III D) holds a common MOTS around both mouths from $t = 18$ (Fig. 6): born while the χ pits are still ∼3.5 coordinate units apart, a tenth out of round (deformation 0.104), with $R = 5.634$, $M_{\rm MS} = 2.817$ and 1.05 times the two throats' summed area. No surface ever closes about either mouth alone: the horizon is born common. At birth the Penrose inequality is violated outright—$R/2 = M_{\rm MS}$ exceeds $M_{\rm ADM}$ by 19.5%—as the phantom permits, since the inequality presupposes the non-negative energy density it lacks [67, 68]: each throat's own Misner–Sharp mass, $R_\star/2 = 1.94$, already exceeds its ADM mass of 1 by the negative energy of the field outside it. From birth the horizon shrinks overall (Sec. VI B). The round star scan, the live monitor, finds the birth 4 units late and 11% small and goes blind while the surface is deformed (Fig. 6(a)): the scan-versus-MOTS offset of Sec. III C.

B. Past the interior failure to a black hole

The merged core is the grid's hard part, not the spacetime's: 21 units after the horizon forms the level-5 segment dies of the censored interior NaN of Sec. VII D ($t = 38.845$), a level-6 segment walks through the failure window, and a level-4 segment carries the remnant to $t = 100$ with no NaN. The finder's history is continuous across both seams (R changes by 0.08% and 0.04% across them).

The remnant is a black hole whose horizon formed around both open throats, and it shrinks as it swallows the phantom field that held them open (Fig. 6(e)): R falls 15.2% from birth—28% of the birth area gone—to $R = 4.7787$ at $t = 100$, rounding from deformation 0.104 to 0.004, while $M_{\rm MS}$ falls from 2.817 to 2.3894. Gravitational waves carry only 4.7% of that loss (the measured $E_{\rm GW} = 8.5 \times 10^{-3}\,M$, Sec. VIII B); the rest is the negative-energy field going down the hole. The shrink is not monotone: on the finder's 84 rows R regrows by up to 0.34%, over $t = 34$–40 and 58–88. The first law of Sec. IV E, measured on this surface, shrinks it under a phantom influx, $T_{ab}k^a k^b < 0$, and grows it through the outgoing shear. Over $t = 51$–60, replayed with every plotfile kept, the observed $\Delta R = -4.1 \times 10^{-3}$ is bracketed by the law evaluated on the found surface: -7.0×10^{-3} from the phantom influx alone, -2.5×10^{-3} with the outgoing shear's growth term added. By $t = 90$–100 the influx is spent: the phantom rate on the surface sits 69-fold below its $t = 45$–55 value, while the radius still falls at 7.7×10^{-4} per unit.

At $t = 100$ the horizon sits 1.4% above $2M_{\rm ADM}$ in radius and 1.4% above $M_{\rm ADM}$ in mass, still shrinking.

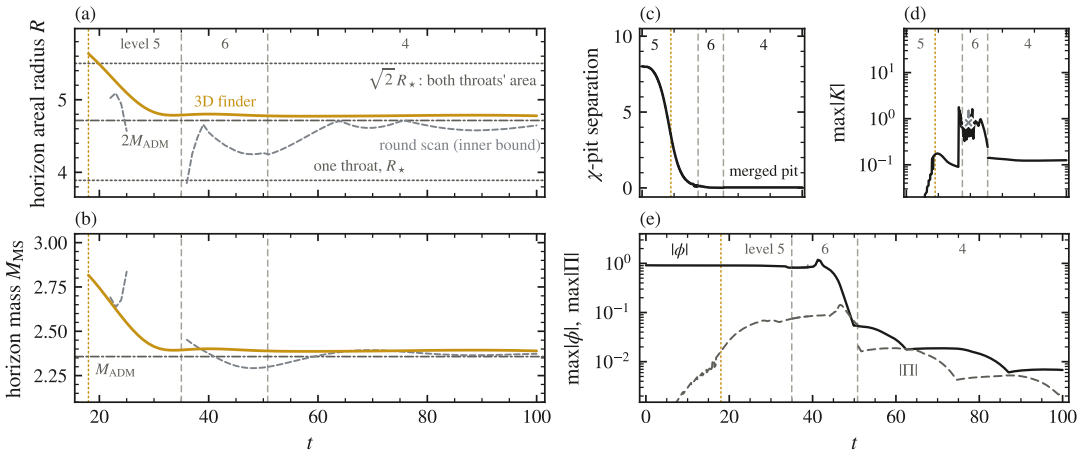


FIG. 6. The head-on merger ($d = 8$ from rest, $p = 0$, opposite signs) on matched, solved data ($M_{\rm ADM} = 2.3573$), and the black hole it makes. One evolution in three segments (faint dashed rules at the seams, $t = 35$ and 50.8; levels named in each panel; Sec. VI B): level 5 to a NaN in h_{11} at the merged core at $t = 38.845$, inside the MOTS (its overrun grey to the death cross in (c)–(e)); level 6 from segment 1's $t = 35$ checkpoint through the failure window; level 4 to $t = 100$ with no NaN. In-code Ψ_4 is identical on both overlaps. (a) Areal radius and (b) Misner–Sharp mass of the common MOTS. Gold: the spectral finder's history (Sec. III D, one row per plotfile, the three replay segments joined), born at $t = 18$ with $R = 5.634$ ($M_{\rm MS} = 2.817$) and shrinking—regrowing by at most 0.34%—toward the Schwarzschild sphere of the pair's mass: $R = 4.7787$ and $M_{\rm MS} = 2.3894$ at $t = 100$, 1.4% above $2M_{\rm ADM}$ and 1.4% above $M_{\rm ADM}$. Grey dashes: the round star scan, the inner bound of Sec. III D, late and small at birth and blind over $t = 26$–35 (Sec. VI A). Rules: one isolated throat, $R_\star = 3.8895$; $\sqrt{2}R_\star$, a sphere with both throats' area; $2M_{\rm ADM}$ ($M_{\rm ADM}$ in (b)). (c) Separation of the χ pits. (d) max $|K|$. (e) Grid maxima of $|\phi|$ (solid) and $|\Pi|$ (dashed): the field is swallowed. Gold dotted rule: horizon birth. Movie: Video 3; youtu.be/3sx9GHRTYVs.

If it settles at the pair's Bondi mass, $R = 2(M_{\rm ADM} - E_{\rm GW} - E_\phi)$ with the measured $E_{\rm GW}$ puts $E_\phi \approx -0.052$, of the sign and order of the flux integrals of Sec. VIII G; a collapsed Ellis–Bronnikov throat in spherical symmetry settles to Schwarzschild on both sides once it has expelled the ghost field [16]. At $t = 100$ the $d = 6$ merger's remnant, still shrinking, lies within 2.2% of the same radius (Sec. VII C), as it should if the end state is set by the pair's mass.

VII. THE ORBITAL CHANNEL

A. Merger or fly-by

Varying the tangential momentum p at $d = 12$ on boosted, solved data separates fly-by from plunge (Videos 4–6, at $p = 0.25$, 0.60 and 0.90: youtu.be/fh3pU7xCU_I, youtu.be/-ejFKpetL1Y, youtu.be/yzvwjC46lJY). The two middle arms are fly-bys: at $p = 0.25$ the pair passes at 2.32 ($t \approx 48$) and has receded to 3.37 by $t = 63.3$; at $p = 0.45$ it passes closer, 2.18 at $t \approx 48$, and recedes to 2.62 by $t = 67.6$. Both reach their stop times with no NaN—the $p = 0.45$ arm with no χ-floor or $\sigma_{\rm KO}$ change to $t = 100$, so a fly-by meets no wall of its own—and the finder, run through each pass, sees no trapped surface on either. The slowest arm, $p = 0.12$, reaches contact (Sec. VII B), and the two faster arms plunge: $p = 0.60$ reaches contact at $t \approx 40$ (pit separation 1.86 at its first segment's stop) and $p = 0.90$ just past its $t = 40$ segment (2.4 and still falling), each collapsing to a merged core on which no common MOTS converges (Sec. VII D). The plunge boundary therefore sits between $p = 0.45$ and 0.60: 90–120% of the circular momentum under the combined point-charge pull, $1 + Q$ times gravity's (Eq. (5)), bracketing the circular value itself. That pull is central: a pair that misses contact keeps its angular momentum and swings past. But from $p = 0.25$ up the approach deepens with speed—closest approach falls from 2.32 through 2.18 to contact, against the Newtonian expectation of a pericentre growing with $J = pd$—so capture here is contact, not an innermost orbit. Nor do the fly-by pairs escape: both start below the circular momentum and remain bound in Newtonian terms, and what follows a pass is set by the throats, which inflate (Sec. VII B).

The recession after the turn is not that of two compact throats: on both fly-bys the mouths inflate, and the constraint norm crosses the campaign's 2.5×10^{-2} criterion as they do, at $t = 63.3$ ($p = 0.25$) and 67.6 ($p = 0.45$)—the trust windows past which nothing is quoted, from the runs or their waves (Sec. VIII D).

B. The race between contact and inflation

At $d = 12$ the finder holds no common horizon at any momentum. The slowest tangential arm, $p = 0.12$, reaches contact at $t \approx 37$ and inflates, and its finder, run from $t = 51$ to the arm's trust window ($t = 56.5$),

holds no common MOTS. In an inward-momentum test run ($p = 0.25$) the mouths inflate faster than the orbit closes—$R_{\min}$ grows +4.7% by $t = 8$ and +72% by $t = 24$ while its infall stalls near separation 2.2—and the finder holds no common MOTS on any of its plotfiles.

The mechanism is the branch selection of Sec. IV C, run by the companion. A lone throat's fate is opposite to a declared kick: pushed outward it collapses, squeezed it inflates. A companion squeezes: each mouth starts in the isolated throat's shape (Table I), yet grows as a lone throat kicked inward on the seed ladder of Sec. IV C, at $\varepsilon_{\rm eff} \approx -0.4\%$ for $d = 12$ at rest and -1.2% for $d = 8$, so every pair's mouths start on the inflation branch, at any momentum, and grow on the throat's own e-fold time $\tau \approx 5.12$ (Sec. IV B). That the solve keeps a declared seed is checked directly: the $\varepsilon = +0.1$ kicked single, run through the constraint solve, is born trapped (a MOTS on its $t = 0$ slice, $R = 4.306$), the fate its kick dictates. A merger is therefore a race against the inflation it starts: the head-on's horizon arrives 3.5 e-folds in and wins; the $d = 6$ merger's arrives at 2.5 e-folds and wins (Sec. VII C); the $d = 12$ approach, 9 e-folds to its closest pass, always loses. The design rule for an off-axis merger follows: short separation and small tangential momentum.

C. The $d = 6$ merger

The design point that wins the race is $d = 6$ with a small tangential twist, $p = 0.10$ per mouth. A level-3 test run merges, and the production chain repeats it on the $L = 128$ box (Fig. 7): the spectral finder holds a common MOTS around both mouths from $t = 13$—2.5 e-folds into the inflation, the pits still 3.6 apart—born with $R = 5.60$ ($M_{\rm MS} = 2.80$, deformation 0.115) and shrinking, with regrowth of at most 0.38% over the finder's 88 rows, by 12.8% to $R = 4.883$, $M_{\rm MS} = 2.441$, deformation 0.016 at $t = 100$, still falling: a black hole within 2.2% of the head-on remnant's radius, reached from an off-axis approach ($p = 0.10$ is about 14% of circular at $d = 6$). As in the head-on, the horizon is born common—no surface ever closes about either mouth—and it swallows the pair mid-inflation.

The chain crosses a harder wall (Sec. VII D) in four segments: level 5 from $t = 0$ to $t = 26.26$; a $\sigma_{\rm KO} = 1.0$ segment to $t = 30.5$; the χ-floor segment ($\min \chi = 10^{-4}$), which sails past every earlier death to $t = 36.7$; and a final level-4 segment, the wall settings reverted, to $t = 100$ with no NaN. The floor acts only inside the horizon, over $t = 30$–36.7; everything quoted comes from the finder's surface outside it.

D. Interior failure after horizon formation

Two failure modes stand between a merged core and its settled remnant, and each has a measured cure. The first is a blow-up of the metric component h_{11} at the merged core's centre: the head-on's level-5 segment at $t = 38.845$, the $d = 6$ chain's restarts at $t = 27.5$–30.5, the $p = 0.60$ plunge at $t = 55.5$—in every case with the global constraint norms clean to the last step. It is not under-resolution: on the $d = 6$ chain the level-6 restart dies at $t = 29.4$ and the level-7 restart dies *earlier*, at $t = 27.5$ with $\max|K|$ still climbing—a finer grid tracks the core's collapse further and meets the failure sooner—and raising the dissipation to $\sigma_{\rm KO} = 1.0$ does not cure it. What cures it is the χ floor: with $\min\chi$ capped at 10^{-4} the $1/\chi$ steepness runaway cannot complete, and both the $d = 6$ chain and the $p = 0.60$ plunge cross their death points and run to $t = 100$, the $d = 6$ final segment then running floor-free. The second mode is the K wall: $\max|K|$ itself running away at the merging core, $93 \to 172$ in one step on the $p = 0.90$ arm at $t = 45.3$. On the $d = 6$ chain level 6 cleared the runaway that stopped level 5; on the $p = 0.90$ arm a level-5 continuation removed the runaway but died at the same instant ($t = 45.28$).

Where a horizon forms, the failure is censored. The core fails only after a common MOTS encloses it: the $d = 6$ cell NaNs arrive 15–18 units after its horizon, the head-on's 21 after its, and lone collapses die behind their horizons at level 3 too ($\varepsilon = +10^{-3}$ at $t = 40.1$, 15 units after its horizon forms; $\varepsilon = +10^{-2}$ dressed with $\varepsilon_2 = 10^{-2}$ at $t = 27.6$, 14 after), while no level-4 lone collapse dies. The failure is local, the exterior is already set, and the finder's surface rides through every crossing unmoved: whatever diverges is the coordinate representation of a merged interior, behind the horizon.

The record thus tests weak cosmic censorship outside the physically reasonable matter it is posed for [84, 85]: in every collapse resolved here—the head-on across its segments, the $d = 6$ chain at levels 5–7, and the lone collapses—the evolution fails, if at all, only inside a MOTS. This is evidence for censorship under null-energy violation, not a proof—the failure is a coordinate blow-up read on the slice, not a located curvature singularity—and the plunges whose horizon never converges (Sec. VII A) leave their failures uncensored on the slice, at the edge of what the grid resolves.

VIII. GRAVITATIONAL WAVES

Every channel radiates; of the energies measured, the horizonless fly-by's at the capture boundary is the largest, the plunge's a floor. The peak $|r\Psi_4|$ of the dominant mode at the innermost sphere is 6.1×10^{-2} for the fly-by, 5.1×10^{-2} for the plunge, 4.8×10^{-2} for the head-on, 2.1×10^{-2} for the $d = 6$ merger, 1.1×10^{-2} for the vacuum reference and 2.9×10^{-3} for the collapsing throat (Fig. 8). Each record propagates outward at $v/c = 0.82$–1.00 between neighbouring spheres (Fig. 8, right; Sec. VIII E). The inner spheres sit in the near zone: the dominant periods are comparable to the extraction radii, and the peak $r\Psi_4$ falls across the spheres by 24% on the head-on and 20% on the fly-by. The radiated energy converges

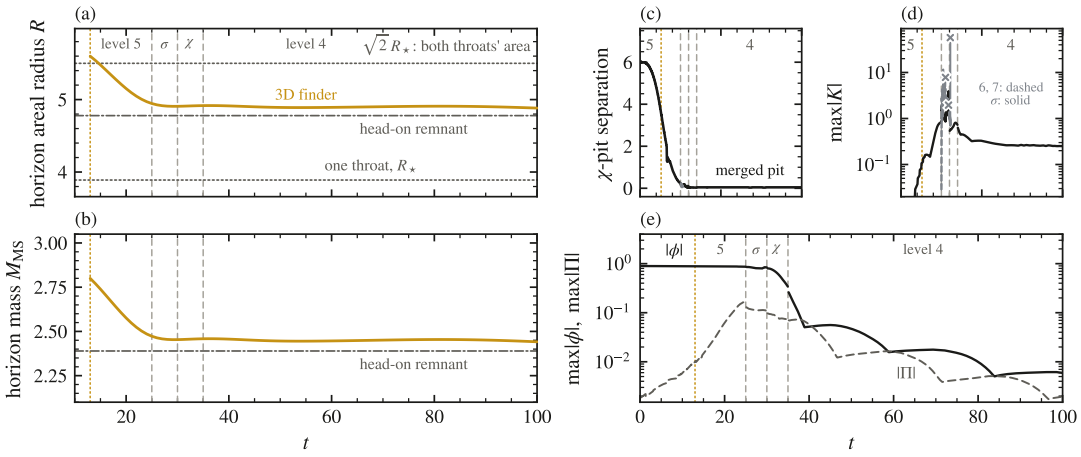


FIG. 7. The off-axis merger ($d = 6$, tangential $p = 0.10$, opposite signs, boosted, constraint-solved data) carried through its merger wall, and the black hole it settles to. One evolution composed across its segments (faint dashed rules at the restarts, $t = 25$, 30 and 35; levels named in the panels, σ and χ for the two wall segments): level 5 from $t = 0$, stopped at $t = 26.26$ as the merged core's K ran away; the $\sigma_{\rm KO} = 1.0$ segment; the χ-floor crossing segment; the final level-4 segment to $t = 100$, no NaN. (a) Areal radius and (b) Misner–Sharp mass of the common MOTS: gold, the spectral finder's history (Sec. III D), born at $t = 13$ and falling from above toward the head-on remnant's values (dash-dotted rules, $R = 4.7787$; $M_{\rm MS} = 2.3894$ in (b)). Dotted rules in (a): one isolated throat, $R_\star$; $\sqrt{2}R_\star$, both throats' area. (c) Separation of the χ pits (grey: segment 1's overrun past its restart checkpoint). (d) The wall: max $|K|$ of every arm—the used segments in ink; the level-6 and level-7 diagnostic restarts (grey dashed) die at $t = 29.4$ and, *earlier*, 27.5, each an h_{11} NaN at the core's centre with the global norms clean to the last step; the σ arm (grey solid) clears the K wall and dies at the same cell at $t = 30.5$ (death crosses); the χ-floor and final segments carry the history through. (e) Grid maxima of $|\phi|$ (solid) and $|\Pi|$ (dashed): the field is swallowed. Gold dotted rule: horizon birth. Movie: Video 7; youtu.be/jNnMAXPjyRA.

outward instead (Fig. 10(c) in Appendix A): on the head-on's seven spheres, out to $R/M = 19$, it holds to 1.5% from $R = 28$ outward and exceeds that by 28% only at $R = 10$, while the vacuum controls' is flat to 5% over their spheres. Every energy is therefore read at the outermost sphere that covers its window, its change from the next sphere in the error bar (Fig. 9(d)); the records drawn in Sec. IX are the innermost sphere's, whose near-zone excess is 13% in rms amplitude, from that 28% energy excess.

Every quoted burst arrives with its physical trigger—collapse, contact or closest pass, tens of units after the first light-crossing $t \approx R$—and the per-sphere envelopes rise with it (Fig. 8, right). The solved data's own transient is small: on every binary the first light-crossing window, $t - R \leq 10$, carries at most 2.7% of the quoted band energy. On the seeded singles the wave is the throat's, not the seed shell's (Sec. VIII A).

Each channel is set against black holes at its own d and p (grey in Fig. 8(b)–(d)). The vacuum merger, at $d = 6$ and $p = 0.10$, peaks at the drainhole merger's retarded time, 3.0× lower. The fly-by's control, at $d = 12$ and $p = 0.45$, is unbound in vacuum (circular 0.204, parabolic 0.289): the black holes start at periapsis and coast apart, radiating $1.05 \times 10^{-3}\,M$ in a single cycle (peak $|r\Psi_4| = 9.0 \times 10^{-3}$, 6.7× below the fly-by's), whereas the drainhole pair falls from 12 to 2.18 under a pull six times gravity's and outradiates it by 73–74×. For scale, the closest non-merging black-hole encounters computed in vacuum radiate $\approx 2 \times 10^{-2}$ of their total mass in all multipoles [69].

A. Collapsing throat

A lone throat radiates only if its collapse is not spherical. The spherically kicked arm emits nothing above the $\ell = 2$ numerical floor, while the same kick with an $\varepsilon_2 = 0.05$ quadrupole produces a burst of peak $|r\Psi_4| = 2.9 \times 10^{-3}$, 44× the floor of the matched level-4 control at $R = 14$, carrying $2.5 \times 10^{-5}\,M$ (integrated to $t = 58$, before the numerical floor dominates). The burst arrives in radius order at 0.8–1.0 c, carries the same $r\Psi_4$ to four spheres within 7.2%, and is linear in the seed: 4.85× for a 5× kick, ratios 1.99/1.97/1.87 at $R = 10/14/18$ for a halved seed, linear to 1–16% across $\varepsilon_2 = 0.005$–0.05. The pure quadrupole—$\varepsilon_2 = 0.01$ with the radial kick removed—radiates the same amplitude to 2%, its waveform leading the kicked arm's by 1.0–2.0 M: the deformation sets the wave, the kick only its clock. The ringdown is read at $\varepsilon_2 = 0.05$: the 0.01 burst, matched to the radial kick, stands only ≈5× above the spherical control's floor at $R = 14$, which the linear scaling meets at $\varepsilon_2 \approx 2\times10^{-3}$. The remnant rings at period 20.6 M (dashed in Fig. 8(a); an independent least-squares fit gives 20.61 ± 0.15), with no drift across the window (half-periods 9.4, 10.3 and 9.0 M), the Schwarzschild fundamental of a black hole of mass 1.226 ± 0.009 [71], near the horizon's late mass

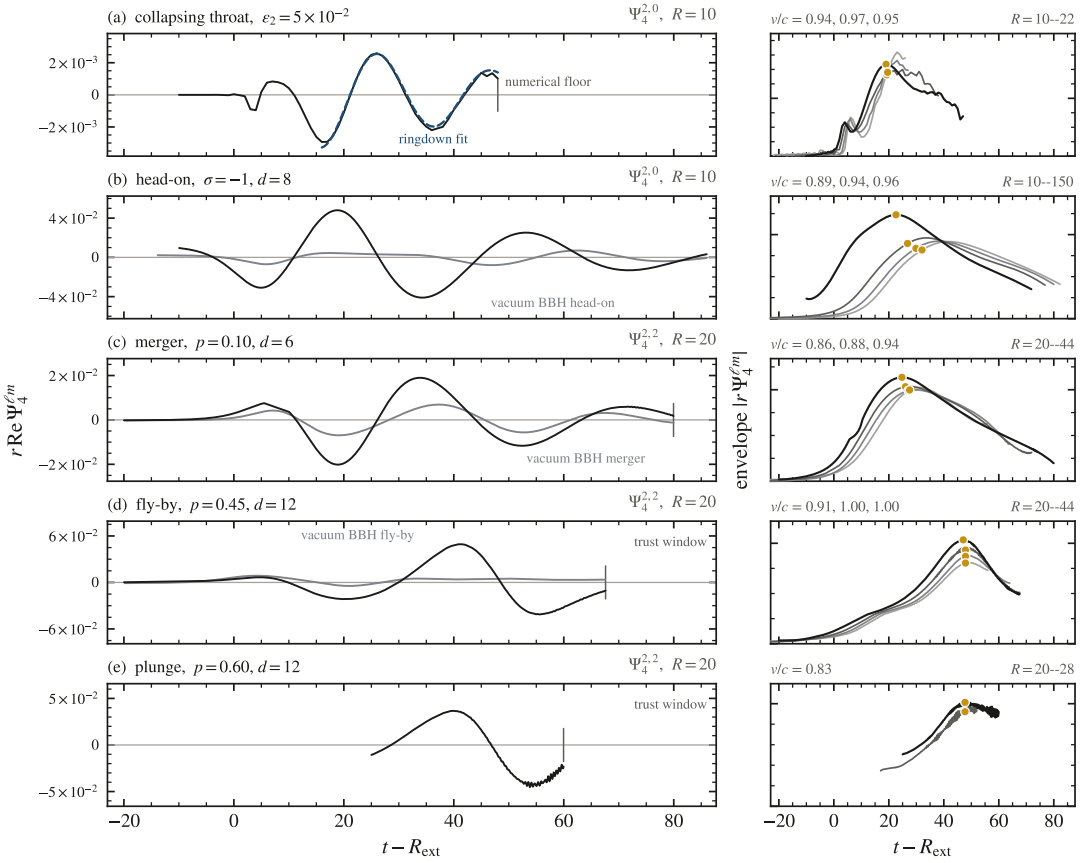


FIG. 8. The waveform gallery: one row per scenario on one retarded clock. Left: $r\,\mathrm{Re}\,\Psi_4$ of the dominant mode at the innermost sphere, each row on its own scale; row (a) is the $\varepsilon_2 = 0.05$ arm, with the ringdown fit of Sec. VIII A dashed (period $20.6\,M$). Caps mark where the drawn record ends: the collapsing throat before its numerical floor exceeds 14% of the burst peak; the fly-by at its trust window ($t - R = 67.6$), the plunge at its own. Row (b) is the same head-on rerun in an $L = 512$ box with spheres to $R = 150$, clean to the clock's edge. The merger runs through remnant ringdown, with one linear bridge across a segment seam. Grey under (b)–(d): the vacuum controls, black holes at the same d and p on bare punctures, on the same scale: the vacuum head-on rings 5.4× lower and crests later (Sec. VIII B); the vacuum merger peaks at the drainhole merger's retarded time, 3.0× lower; the vacuum fly-by radiates one cycle as the holes swing away from periapsis, 6.7× below the fly-by's arch. Right: the amplitude envelope at every sphere on its own retarded time (ink to faint, inner to outer)—the upper envelope of $|r\Psi_4|$ for the complex $(2,2)$ modes, the analytic-signal envelope for the real $(2,0)$ modes—which folds onto one curve for a signal moving at c and falling as $1/r$. Gold dots: the correlated wavefront. The quoted v/c is the whole-waveform lag between neighbouring spheres (Sec. VIII E), the plunge's the burst-peak lag: throat 0.94/0.97/0.95, head-on 0.89/0.94/0.96, merger 0.86/0.88/0.94, fly-by 0.91/1.00/1.00, plunge 0.83. Movies of the far-zone head-on, whose remnant spreads with the coordinates while its area holds, and of the vacuum head-on and fly-by: Videos 11 youtu.be/vesOxaP5GSg, 8 youtu.be/aRWvD4W8OOg and 9 youtu.be/65_3kw4obvE.

(1.16–1.21, periods 19.6–20.4 M) and not the larger mass it carries at the window's start (1.47, period 24.7 M).

B. Head-on ringdown

The head-on's signal is a bell with no chirp: the $(2,0)$ breathing mode at $R = 10$ swings through 4.8×10^{-2}, -4.1×10^{-2}, 2.5×10^{-2} at $t = 29$, 44, 63—the half-period stretching from 15.5 to 18.8, the amplitude falling ×0.61–0.86 per half-swing—and arrives ∼4.4 units later per 4 units of extraction radius ($0.92\,c$), outgoing across all three spheres. On the detector track it is the bell that slows from 707 to 410 Hz in Fig. 9(c). Band-integrated about its peak, the bell carries $E_{\mathrm{GW}}/M = 8.5\times10^{-3}$ at the outermost sphere (1.1×10^{-2} at the innermost, its near-zone excess). The vacuum control—two equal black holes released from rest at the same $d = 8$—rings the same $(2,0)$ bell 14× quieter ($E_{\mathrm{GW}}/M = 6.0\times10^{-4}$ at its outermost sphere, against the published 5.5×10^{-4} for infall from rest at infinity [78]) and 28 units later at $R = 14$ (its crest at $t = 61$ against 33): without the scalar's pull the holes fall slower and hit softer.

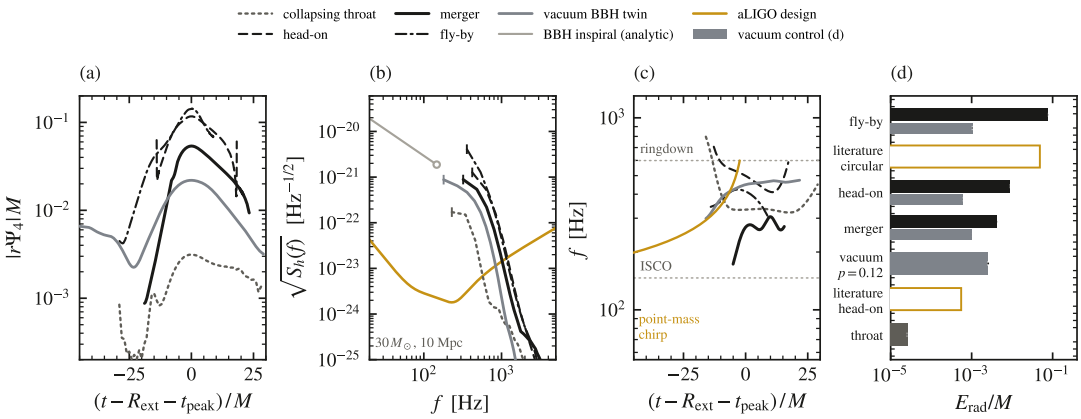


FIG. 9. Every source at one calibration: total mass $30\,M_\odot$ at 10 Mpc, each record in units of its own total mass; the vacuum twin is two black holes with the separation and momentum of the $d = 12$, $p = 0.12$ arm (Sec. VII A). (a) Upper envelopes of $|r\Psi_4|M$ at the innermost sphere, aligned at each peak: fly-by loudest (1.4×10^{-1}) to lone throat (3.1×10^{-3}). (b) Strain amplitude spectral densities over the aLIGO design floor (gold), each from its own low-frequency corner (one cycle per record, ticked). The curves are falling power laws (slopes -4.0 to -5.8) with no resolved maximum, so any "peak" read here is the corner $1/T$. Faint line: the Newtonian inspiral of the same binary, $|\tilde{h}| \propto f^{-7/6}$ to its ISCO at 147 Hz. (c) Instantaneous frequency against the point-mass chirp (gold), the ISCO (147 Hz) and the equal-mass Kerr ringdown (599 Hz): the vacuum twin climbs the chirp and peaks at $fM = 0.065$; no wormhole channel does (Sec. IX A). (d) Energy radiated in the dominant multipole at the outermost sphere covering each record's window, the tick its change from the next sphere in; grey under each source, its vacuum control at the same d and p. Open gold bars: published equal-mass vacuum values: head-on infall from rest at infinity, 5.5×10^{-4} [78]; quasi-circular inspiral from infinite separation and merger, 4.84×10^{-2} [72]; all multipoles. Energies are band integrals of $|\int r\Psi_4\,\mathrm{d}t'|^2$ over one retarded window per record, ending at its trust or noise gate (the lone throat's at $t = 58$, before its floor), at the spheres whose record covers it (the fly-by's inner two), doubled for $\pm m$; gravitational channel only. Movie of the vacuum twin: Video 10; youtu.be/NkcpGiJIxXI.

C. Merger burst

The $d = 6$ merger's $(2, 2)$ burst is launched at contact—the common MOTS of $t = 13$ (Sec. VII C)—peaks at $|r\Psi_4| = 2.1 \times 10^{-2}$ on $R = 20$ at $t \approx 39$, folds onto one curve across $R = 20$–44, and the remnant's ringdown carries the record, clean, to $t = 100$ on the final segment (Fig. 8(c)).

D. Fly-by and plunge bursts

Each orbital arm emits one $(2, 2)$ burst at its closest pass or plunge, with no ringdown on the fly-bys—no remnant forms. Through the outermost sphere that covers each burst ($R = 28$; the plunge's, $R = 20$), $E_{\rm GW}/M = 3.53 \times 10^{-2}$ at $p = 0.25$, 7.6×10^{-2} at $p = 0.45$ (damped run; the undamped control covers only $R \lesssim 34$ to $t = 70$, Sec. III B), and at least 4.1×10^{-2} at $p = 0.60$, each in its own pair's measured mass (Sec. II E). Of the measured energies the largest is at the capture boundary (Sec. VII A), where the pass is deepest and slowest: the $p = 0.45$ arch carries 2.2× the shallower fly-by's energy and 1.9× the plunge's floor. Each fly-by's arch is integrated to its run's trust window ($t - R \leq 63.3$ and 67.6); the next sphere in reads 3.63×10^{-2} and 7.8×10^{-2}. The plunge's burst is cut by its run's trust window, 12 units past its $R = 20$ peak, so its energy is a floor and the boundary-to-plunge ratio a ceiling.

E. Vacuum twin

The chirping reference—a vacuum binary at $d = 12$, $p = 0.12$, equal masses, no scalar—behaves as a binary black hole must: it merges, climbs the point-mass chirp 299 → 473 Hz and turns toward the equal-mass Kerr mode (Sec. IX A). It also calibrates the wavefront method: its flat-topped merger envelope defeats peak timing, which reads $0.57c$ on it against the $0.82c$ of its whole-waveform lag, so every speed here but the plunge's is a correlated lag.

F. Scalar channel and its sign

Ψ_4 is blind to a spin-0 wave. Two opposite scalar charges—every merging pair here (Sec. V)—carry a dipole, where a vacuum binary's leading moment is the quadrupole. On every binary the scalar stream rides the offline Ψ_4 spheres (Sec. III C), so the two channels compare sphere by sphere (Fig. 12 in Appendix B). The recorded kinematic flux $F_{\rm kin}$ approximates the scalar energy crossing S_R per coordinate time: the flux through the world tube of S_R of the canonical energy current

$J^\mu \equiv -T^{(\phi)\mu}{}_\nu\, t^\nu$, $t^\mu = \alpha n^\mu + \beta^\mu$ [46, 47],

$$F_\phi^{\rm geo} = \oint_{S_R} \alpha\, s_i J^i \sqrt{\sigma}\, \mathrm{d}^2x = -\oint_{S_R} \alpha\, \partial_t\phi\; s^j \partial_j\phi\, \sqrt{\sigma}\, \mathrm{d}^2x \;+\; O(\beta), \tag{13}$$

positive for an outgoing canonical wave (s^i the sphere's outward unit normal, $\sqrt{\sigma}$ its area element, $\partial_t\phi = \alpha\Pi + \beta^i\partial_i\phi$). At $R = 30$ the background's area mean of $\alpha^2\chi^{-1/2}$ is ≈ 0.94, so $F_{\rm kin}$ overstates the geometric flux by about 6%, shift terms neglected.

The channel is dipole-dominated and comparable to the gravitational one. On the fly-by at $R = 30$, $\ell = 1$ carries the scalar power to one part in $10^{6.2}$—$\ell = 0$ and $\ell = 2$ are forbidden by symmetry for $\sigma = -1$, so the remainder measures numerical symmetry breaking—and leaves as a single outward arch, complete within the sphere's trusted record (Fig. 12(a)). Over that whole record $|E_\phi| = 4.7\times10^{-2}\,M$, 0.56 times the gravitational energy that crosses the sphere with it (0.61 band-limited, 0.60 by the wave-zone mode sum); mid-burst, at $t = 60$, the ratios read 2.4 and 3.7, the inner sphere 0.6 (Fig. 12(c)). Part of that energy is not radiation: before the arch $F_{\rm kin}$ rises on every sphere at once, ahead of any signal from the throats, falling with radius as $R^{-2.5}$—the moving charges' near field, 14% of $|E_\phi|$; the arch alone gives $|E_\phi|/E_{\rm GW} = 0.48$. Downstream we use only the channel's multipole, direction, sign and order of magnitude.

Gravity couples to minus the canonical stress tensor, Eq. (1), so the physical energy flux is $-F_{\rm kin}$: the outgoing $\ell = 1$ wave—on the fly-by and, once their horizons form, on the head-on and the $d = 6$ merger—carries negative energy. If the ADM energy balance holds with this term, a bound pair emitting it gains energy and radiation reaction is *anti-damped*: the classical face of the standard objection to ghosts [62, 86]. Classically these binaries make the same trade, and the head-on remnant approaches the pair's Bondi mass, a little above $M_{\rm ADM}$ and still shrinking at $t = 100$ (Sec. VI B). The dipole also returns angular momentum—through both spheres its physical flux is opposite to the gravitational waves', which carry off the orbit's sign—and it does not invert the nonlinear memory [90–92]: per unit energy an equatorial dipole projects onto the $(2,0)$ memory with the opposite sign to a quadrupole, so the ghost's negative flux adds to the gravitational memory, by 0.34–0.46 of it on the fly-by.

G. Horizon censorship of the scalar channel

The head-on's chain carries the scalar stream end to end, and its horizon switches the channel off (Fig. 12 in Appendix B). After the common MOTS closes ($t = 18$), the mouths' remaining hair leaves as a single $\ell = 1$ pulse along the collision axis, cresting later at each sphere in causal order ($t = 34$ and 38 at $R = 14$ and 18), and the envelope then decays: log-linear fits over $t = 30$–80 give $\tau =$ 22/38/73 at $R = 10/14/18$. Underneath, the remnant rings down its hair: fitted from $t = 40$ as one damped oscillation, the dipole at $R = 10$ rings at $\omega = 0.123$ and decays at 0.037, against 0.1226 and 0.0409 for the $\ell = 1$ scalar quasinormal mode of a Schwarzschild hole of the final MOTS mass ($M\omega = 0.2929 - 0.0977\,i$ [71]): the frequency agrees to 1% for fits started anywhere over $t = 35$–45, and the damping is 10% slower. Integrated from where each sphere's flux turns outgoing ($t = 26$ at $R = 10$; before it the open mouths' static hair drains canonically inward) to $t = 80$, the energies keep the sign of Sec. VIII F, $E_\phi = -0.0500/-0.0571/-0.0594$ in units of m.

The $d = 6$ merger, the one other encounter with a horizon (common MOTS from $t = 13$; Sec. VII), sheds the same way on its own glued record: its dipole crests at $t = 38$ on $R = 20$ and falls exponentially, $\tau = 22$ (fit over $t = 45$–95), 19× down at $R = 14$ by the record's end. Without a horizon there is nothing to switch the channel off: the fly-by radiates its arch as the mouths swing past one another, and what ends that emission is the encounter itself—the mouths leave intact, still carrying the charges that source the dipole.

The lone throat shows the same contrast. Re-run at level 4 with the scalar stream, the $\varepsilon = +10^{-2}$ throat collapses on schedule—a permanent MOTS from $t = 11$, R falling $3.88 \to 2.33$ at its floor—and its scalar envelope, after the burst, falls across the spheres by $t = 80$, with late e-fold times of 6–7. The unkicked level-4 throat, which inflates with no trapped surface by scan or flow finder, instead grows quasi-exponentially in the $\ell = 0$ monopole (e-fold ≈4.6, shortening across $t = 45$–75), as in a box-doubled ($L = 128$) copy, to 3% over $t = 50$–80, so the growth is the throat's and not the boundary's or the sponge's; beyond $t \approx 80$ the constraint norms grow, and only $t \lesssim 80$ is counted. The contrast is confounded with the fate: the horizonless arms are also those whose mouths inflate toward the spheres. What is measured is that the scalar channel shuts off behind a horizon and grows while a throat inflates.

IX. OBSERVATIONAL SIGNATURES

A. The frequency track

Morphology alone does not mark a drainhole burst: cut to the signal's duration and scaled to 60–300 $M_\odot$, binary-black-hole templates (`IMRPhenomD` [94, 95]) match the bursts in the aLIGO design noise above 20 Hz [96] with fitting factors of 0.893–0.995, as they match the vacuum control (0.935–0.951), on truncated near-zone records. The frequency track separates a drainhole burst from a binary black hole by what comes before the ringdown (Fig. 9(c)). The vacuum twin peaks at $fM = 0.065$ and climbs the point-mass track before turning toward the equal-mass Kerr mode at $fM = 0.0885$ [70, 71]: a monotone rise, then a fixed ringdown frequency. No drain-

hole channel follows it. The head-on's frequency *falls*, by $\times 0.58$, from contact onto the $\ell = 2$ quasinormal mode of a Schwarzschild hole of its final MOTS mass, 397 Hz at this calibration, which it reaches to 3% at the track's minimum, before a late upturn: the remnant does not spin, and the shrinking of its horizon would raise that mode, not lower it. The fly-by's frequency falls as the pair separates; the merger's rises after contact; the lone collapse rings at its own Schwarzschild mode (Sec. VIII A), 0.56 of the equal-mass Kerr one. Wherever a remnant's ringdown is resolved—the lone throat's, the head-on's and the head-on's scalar dipole (Sec. VIII G)—it is the ordinary quasinormal spectrum of the black hole left behind, as in any catalogue event [32]; the wormhole is in the burst before it.

B. Energy per event

The detector-independent deliverable is the energy per event in units of the source's ADM mass M, in the dominant multipole about each source's axis (change from the next sphere in, in parentheses): fly-by $7.6 \times 10^{-2}\, M$ (1.9%), merger $4.2{\times}10^{-3}\, M$ (1.1%), head-on $8.5{\times}10^{-3}\, M$ (1.4%), collapsing throat $2.5 \times 10^{-5}\, M$ (3.6%), vacuum twin $2.4 \times 10^{-3}\, M$ (10%). Since $\Omega_{\rm GW}$ scales as event density times energy per event, these convert any bound on $\Omega_{\rm GW}$ [28, 29] into a bound on the abundance of primordial drainhole encounters, provided the ghost scalar exists at that epoch, as they do the rate-density limits of the LVK O3 all-sky burst search [73]. The scalar channel adds comparable energy of opposite sign (Sec. VIII F).

X. DISCUSSION

All four fates of Sec. I occur, and the fast plunge, each with a measured selection rule: the relative scalar sign separates scattering from merger (Sec. V), the seed's sign separates collapse from inflation (Sec. IV), and the orbital momentum separates plunge from fly-by at 90–120% of circular (Sec. VII). For detection, three results combine: no inspiral exists (Sec. X A), the wormhole phase is a burst with no chirp ahead of it, its frequency falling or drifting until the remnant's ordinary ringdown takes over (Sec. IX A), and the scalar counterpart decays once a horizon forms (Sec. VIII G). A surviving wormhole signature is a drifting-frequency burst with a scalar counterpart; a settled ringdown without one is already the descendant.

A. No wormhole inspiral

A drainhole binary cannot radiate an inspiral. Each throat's unstable mode gives it a lifetime $t \approx \tau \ln(1/\varepsilon)$, with $\tau = 5.1$–5.2 in isolation. An inspiral needs many orbits. Under the combined pull of Eq. (5), six times gravity at large separation and softened as $(d{+}\delta)^{-2}$ at the separations evolved (Sec. V B), a circular orbit at $d = 12$ has period $P \approx 75$–93. No pair starts quiet enough to wait that long: the companion itself plants the seed—the effective inward kick of 0.4% at $d = 12$ (Sec. VII B)—an ε worth $t \approx \tau \ln(1/\varepsilon) \approx 28$–$29$ units—under half a period. Even $\varepsilon = 10^{-15}$ buys only 1.9–2.4 periods, and each further factor of ten adds only $\tau \ln 10 \approx 12$ units. The orbit's demand grows with separation as $P \propto d^{3/2}$, while the throat's endurance grows only as $\ln(1/\varepsilon)$, and the companion's kick is part of the binary itself.

Radiation reaction cannot close the gap either: the leading radiative moment, the scalar dipole, carries negative energy comparable to the gravitational channel's (Sec. VIII F), which, if the ADM balance holds, pumps the orbit rather than draining it. The encounters evolved here are not secular: each is settled within one passage by the conservative pull and contact (Sec. VII A), with both channels evolved self-consistently, and over the fly-by the scalar carries only 0.56 of the gravitational energy (Sec. VIII F), so a close pass loses energy on net. Pumping concerns the slow, wide orbits an inspiral needs, where the dipole, the lower multipole, dominates—orbits no throat survives to reach.

The wormhole-specific signal is therefore a short transient: within a few tens of M each mouth collapses or inflates, and the plunge, burst and conversion of Secs. VI–VIII are all that this source class emits *as wormholes*; any long chirp belongs to the objects the throats become. The argument holds while the throat's unstable mode exists, for the minimally coupled ghost scalar without spin of the González–Guzmán–Sarbach analysis. Rotation can remove that mode—in five dimensions above a critical angular momentum [49], in four as perturbative analyses indicate [50–52], with spinning Ellis wormholes existing up to an extremal limit [48]—and other supports are radially stable [53, 54], so a spinning or differently supported pair that outlives an inspiral is not excluded here. The same lifetimes rule out a static ghost-supported throat as a long-lived black-hole mimicker of the kind ringdown and echo tests target [87, 88].

B. The inflating half of the population

Collapse is half the census. The branch is chosen by the perturbation's sign (Sec. IV C), so a sign-symmetric seed distribution converts about half of a population of lone throats and sets the other half inflating. A companion tilts the census toward inflation: its squeeze starts both mouths on the inflation branch (Sec. VII B), and only pairs whose common horizon wins the race convert—here the head-on and the $d = 6$ merger, no $d = 12$ pair. An inflating throat creates, from regular data, an expanding anti-trapped region (Sec. IV F), which under the null energy condition would need an initial singularity [59, 60]; unlike the false-vacuum bubbles of those references and the domain-wall wormholes of Ref. [6], whose inflating interiors pinch off behind a horizon, it lies on our side of

the throat and grows into the surrounding space, as the massless throat's explosion does [13]. Over its record the neck accumulates 1.3 areal e-folds and the boundary 3.0, against the ∼60 of cosmic inflation [61]: the record shows the mechanism, not the magnitude.

C. Topological censorship

Topological censorship forbids an observer at infinity from probing the topology of a globally hyperbolic, asymptotically flat spacetime that satisfies the null energy condition [89]. A traversable wormhole evades it only by violating that condition, and the two branches end the evasion in opposite ways. On the collapse branch, alone or in a merger, the throats end inside a MOTS while the phantom outside is swallowed or radiated away (Secs. IV E, VI B and VIII G): the wormholes are not removed—no classical evolution changes the number of asymptotic ends [19]—but end hidden behind a horizon, as the theorem would have them. On the inflation branch no trapped surface forms and the throat stays open (Sec. IV F): the evasion persists, and a distant observer meets the expanding anti-trapped region of Sec. X B—ghost field, whose energy density is nowhere positive—with no horizon in front of it over the whole record. The far universes enter only as compactified punctures; fluxes through the throats into them are not measured.

D. Scope

The results hold for one matter model and one family of data: a single minimally coupled ghost scalar, equal masses, one throat compactness ($m/a = 1/2$) and no spin, on constraint-solved, matched initial data whose momentum is the exact boost (Secs. II D and II E). Horizons here are MOTSs on the evolved slices, which for matter violating the null energy condition need not lie inside event horizons (Sec. III D). Within this family the instability itself excludes survival from any early formation epoch to late times, whatever the tuning; a long-lived relic requires changing the object—spin, which perturbative analyses indicate can remove the unstable mode, or a radially stable support (Sec. X A)—and for either our numbers no longer apply. Nonlinear evolutions of spinning throats are the natural next step.

XI. REPRODUCIBILITY AND COST

Every evolution in Table III is one line in a registry shipped with the results pack—its single knob against a named reference, its outcome, its caveats—and every figure is regenerated by a script in the pack from the archived diagnostic streams. The code, the analysis and the results pack are in the `GRTeclyn` [41] fork at https://github.com/Nikchik-coder/GRTeclyn (tag `v1.0.1-wormhole-merger`, archived at https://doi.org/10.5281/zenodo.23257999; results pack at https://github.com/Nikchik-coder/GRTeclyn/tree/v1.0.1-wormhole-merger/results/merger): the drainhole initial data with its constraint solve and exact boost, the seeds of Sec. II F, the scalar damping and the in-code diagnostics in the evolution code; the spectral MOTS finder, the orientation-corrected surface scan, the figure scripts and a matched-filter pipeline for public strain [30, 31] in the analysis layer; and the parameter files and thinned diagnostic streams of every run in the pack. A table in the pack lists the binaries the runs name and, where known, each one's source commit. The evolutions are bit-reproducible on the production hardware: a lone-throat arm rerun with checkpoint output gives byte-identical physics streams over 633 output rows.

Every evolution ran on one NVIDIA H100 80 GB device, with `GRTeclyn` built CUDA-native against AMReX 26.02. At $\Delta t = 0.02\,\Delta x$ a production arm ($L = 64$, 128^3 base, three refinement levels) advances 4–18 code units per hour, and a $t = 100$ single-throat evolution takes 5.4–5.5 h. Depth is the expense: level-5 arms run at 2–4 units per hour on the full box (the octant $L = 512$ arm, an eighth of the cells, at 28–74, 11.9 h in all)—29.3 h for the head-on's three-segment $t = 0 \to 100$ chain, 46.3 h for the $p = 0.25$ fly-by at level 5, which peaks at 58 GiB of the 80 GB device, and 26.9 h for the $d = 6$ merger chain. The 103 evolutions total 592 GPU-hours, each run's cost recomputed from its archived logs: the logged speed and time span, a restarted segment dated from its checkpoint.

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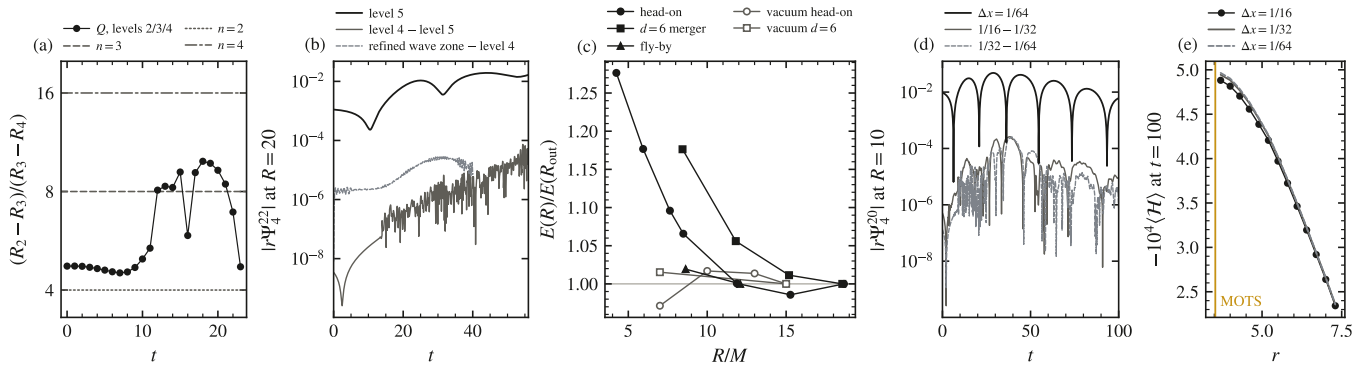


FIG. 10. Convergence. (a) The static throat from exact data at levels 2–4: the three-level factor $Q = (R_2 - R_3)/(R_3 - R_4)$ of its minimal areal radius against the orders $n = 2$, 3, 4 (rules); the dip at $t = 16$ is one level-2 dump. (b) Bowen–York, not production, data: $|r\Psi_4^{22}|$ at $R = 20$ of the $d = 12$, $p = 0.12$ pair at throat level 5, its difference from level 4, and the difference made by refining the wave zone to half the base cell. (c) Radiated energy on each sphere over the outermost sphere's, against R/M: the head-on, the $d = 6$ merger and the fly-by (ink) and two vacuum controls (grey). (d) The far-zone head-on of Fig. 8(b) at finest $\Delta x = 1/16$, 1/32 and 1/64: $|r\Psi_4^{20}|$ at $R = 10$ and each neighbouring pair's difference, peaking at 0.55% and 0.54% of the peak. All three form the common MOTS at $t = 18$, its R and $M_{\rm MS}$ equal between neighbouring levels to at most 0.08%, and the energy at $R = 10$ converges at order 1.7: differences 0.3%, then 0.09%. (e) The same three runs' $t = 100$ checkpoints: shell averages of $\mathcal{H}_{\rm ADM}$ against r outside the common MOTS (gold rule, its largest coordinate extent), read on the shared level-4 grid; each halving of the interior cell moves them by 1.4%, then 0.4%. The outcome: orders 3.0–3.3 (static throat; 2.2–2.3 while the error of reading the exact data dominates), 1.7 (head-on energy) and 1.7 ($\mathcal{H}_{\rm ADM}$); the Bowen–York binary's waveform is resolved to 0.4% of its peak, the energy converges in extraction radius and in resolution, and the interior's constraint violation does not leak past the horizon.

Appendix A: Convergence and Code Health

Figure 10 tests convergence. The static throat, the one configuration with an exact answer, gives at levels 2–4 a three-level order of 2.2–2.3 while the error of reading the exact data dominates and 3.0–3.3 once the evolution's own error does, between the fourth order of the stencils and the second of the time interpolation at refinement boundaries; later the unstable mode amplifies each level's own seed, which is why the unkicked fates differ (Sec. IV C). A pair's branch is set by its data instead: the companion squeezes each mouth like a declared inward kick of 0.4% at $d = 12$ and 1.2% at $d = 8$ (Sec. VII B), while a kick of 10^{-3} already overrides the level-3 seed and a kick's branch survives refinement to level 4 (Sec. IV C). The Bowen–York $d = 12$ binary of Fig. 10(b) at throat levels 4 and 5 agrees in $r\Psi_4$ to 0.4% of its burst peak, and refining its wave zone moves it by 0.15%.

Figure 11 gives the constraint record of every run the main-text figures draw. Its norms are volume averages on a base grid that does not resolve the throats (Sec. III C), so the throats set their absolute scale—four base-grid cells beside the moving pits hold 99.99% of a spike's norm—and the norms serve as monitors, not as wave-zone error bars; at every recorded death the norm sits at or below its earlier value. Whether the interior reaches the waves is tested by the far-zone set of Fig. 10(d),(e): three runs identical but for the throats' finest cell (all at $\chi \geq 10^{-4}$ and $\sigma_{\rm KO} = 0.3$, unlike the production head-on; Table IV) agree pairwise, to the caption's sub-percent figures, in the exterior waveform, the horizon track, the radiated energy and the $t = 100$ Hamiltonian outside the MOTS, so what the interior does at its own resolution stays interior.

Appendix B: Supplementary Measurements

Figure 12 carries the scalar channel's energy, multipoles and validity window (Sec. VIII F) and the horizon's shut-off of the channel (Sec. VIII G).

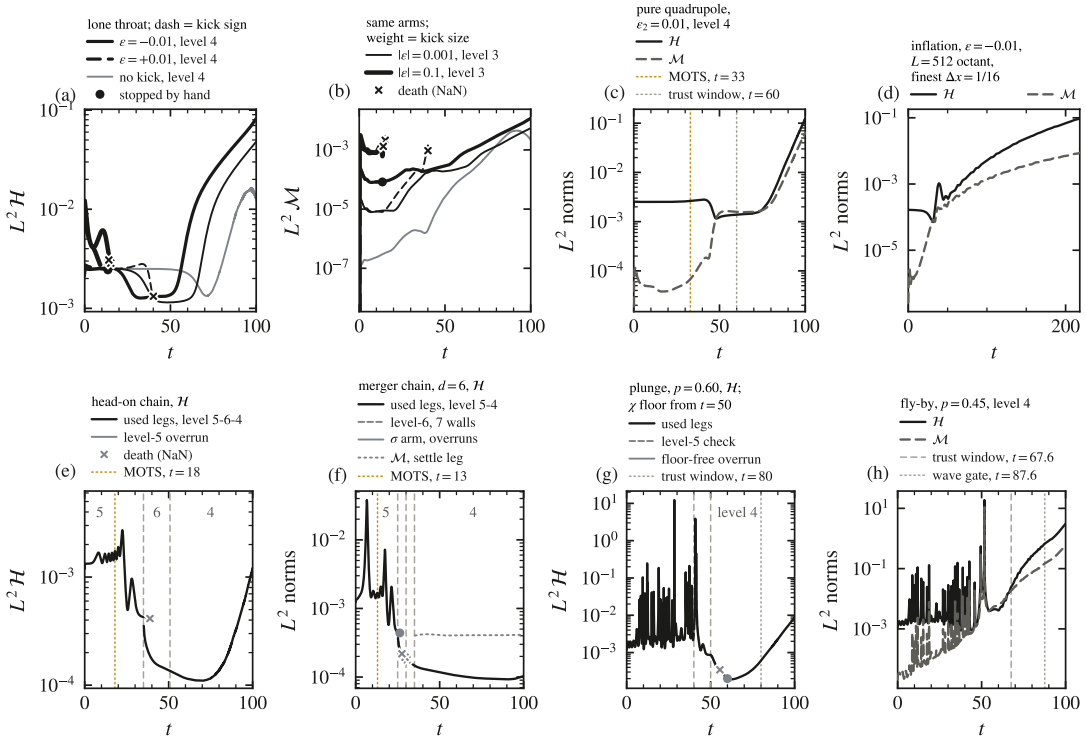


FIG. 11. Constraint evolution of every run the main-text figures draw, each at the highest refinement level it was run at (named in each panel's key): L^2 norms of the Hamiltonian ($\mathcal{H}$, ink) and momentum ($\mathcal{M}$, grey dashed) constraints, box averages on the base grid ($\Delta x = 0.5$; 2 on the octant of (d)) that compare only within one box (Sec. III C). (a),(b) The lone throat of Fig. 3: level 4 for the unkicked throat and $\varepsilon = \pm 10^{-2}$ ($+10^{-2}$ stopped by hand, dot), level 3 for $\pm 10^{-3}$ and ± 0.1 (dash = sign, weight = amplitude, grey = no kick, crosses = deaths). At $t = 0$ the norms carry the exact data's discretisation floor plus the kick's own defect, $0.090\,|\varepsilon|$ (Sec. II F), which feeds $\mathcal{M}$ within the first steps; the $|\varepsilon| \le 10^{-2}$ arms then follow the unkicked norms to 5.4% until the fates part, while $\varepsilon = \pm 0.1$ starts 3–6× high and dies in a one-step overflow at the compactified origin. (c) The pure-quadrupole collapse of Fig. 4 (gold, its MOTS from $t = 33$; dotted, its trust window, $t = 60$): the late rise comes after it. (d) The inflating throat of Fig. 4 to $t = 218$. (e) The head-on chain of Fig. 6 with its ladder (era rules at the restarts, levels per era): the level-5 segment's overrun dies of the censored interior NaN at $t = 38.845$ (cross) with its norm below its value at formation (gold rule, $t = 18$). (f) The $d = 6$ merger chain of Fig. 7: the used segments in ink, every failed arm in grey to its death cross (the level-6 and level-7 restarts dashed; times in Fig. 7(d)). At every end, hand stop and death alike, the Hamiltonian norm sits at or below its start: each failure is the interior NaN of Sec. VII D, not a constraint blow-up. (g) The $p = 0.60$ plunge, level 4 throughout: the floor-free extension (grey) dies of the merged core's cell NaN at $t = 55.5$ with its norm at its floor, the χ-floor segment carries the record to $t = 100$, and a level-5 restart (grey dashed) crosses the merger with clean norms; dotted, the trust window ($t = 80$). (h) The $p = 0.45$ fly-by: the norms sit on their floor through the pass and grow as the mouths inflate; rules, the trust window ($t = 67.6$) and the energy integral's end ($t = 87.6$). The spikes on (g), (h) and the $p = 0.25$ arm, never on the head-on or $d = 6$ chains, are the base-grid norm reading the moving throats' χ pits: on the $p = 0.25$ verification pair four cells next to the pits carry 99.99% of a spike while the level-3 solution is clean, and neither $\max|K|$ nor the waves carry a counterpart.

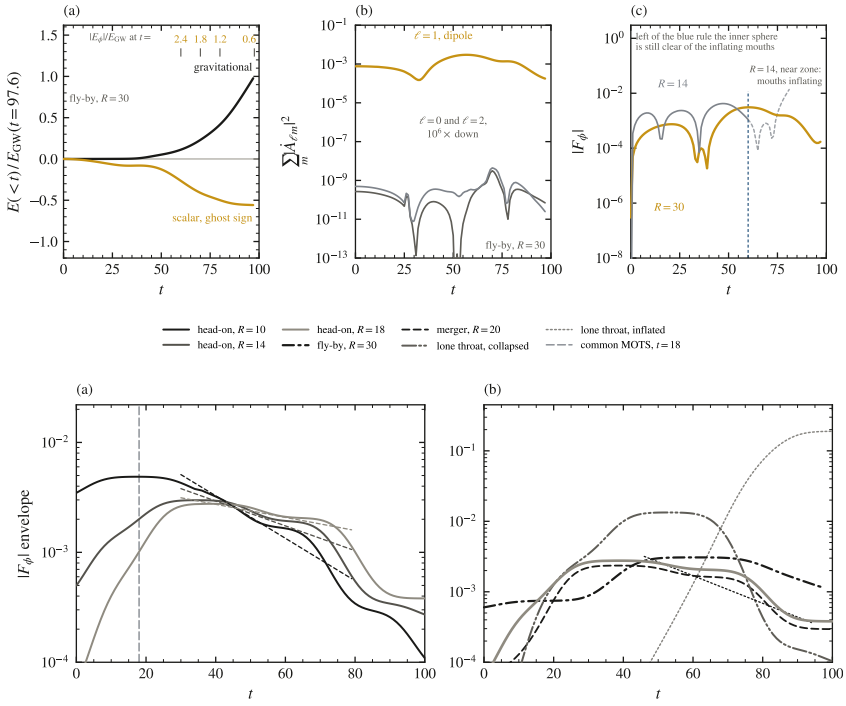


FIG. 12. *Top (channel):* The scalar channel on the Ψ_4 spheres. (a) Fly-by energy through $R = 30$ in units of the gravitational energy crossed by the end of the sphere's trusted record ($t = 97.6$, $t - R = 67.6$): gravitational (ink) and scalar with its physical sign (gold, Sec. VIII F). The ratio $|E_\phi|/E_{\rm GW}$ printed at each cut falls from 2.4 at $t = 60$ to 0.56 over the whole record, as the gravitational burst arrives after the scalar one (Sec. VIII F). (b) Multipole content, fly-by: $\ell = 1$ carries all but $10^{-6.2}$ of the power; $\ell = 0$ and $\ell = 2$, forbidden by symmetry, measure numerical symmetry breaking. (c) Validity window: $|F_\phi|$ at both spheres, each drawn to its retarded gate $t - R = 67.6$. Past the blue rule ($t = 60$) the inner sphere reads the inflating mouths' near field—its climb there has no counterpart at $R = 30$ one light-crossing later—so that stretch is dashed and carries no claim; the $t = 60$ window at $R = 14$ gives 0.6 in place of 2.4. *Bottom (censorship):* Envelope of the scalar flux $|F_\phi|$, horizoned against horizonless arms: a running maximum over 25 units, smoothed in log amplitude, since the raw flux changes sign every ∼13 units. (a) The head-on chain's glued record at $R = 10/14/18$. The common MOTS closes at $t = 18$ (dotted); the remaining hair leaves as one $\ell = 1$ pulse, cresting later at each sphere, then decays (dashed: fits over $t = 30$–80, $\tau = 22/38/73$). The rise before the rule is the open mouths' static hair, canonically ingoing; only the outgoing window is integrated. (b) One sphere per fate (head-on and lone throat $R = 18$, merger $R = 20$, fly-by $R = 30$): the merger, behind its horizon from $t = 13$, decays like the head-on ($\tau = 22$, dotted fit); the fly-by, horizonless, shows its burst's complete arch, drawn to $t - R = 67.6$ (its trust window). Mid-grey: the lone throat twice—the $\varepsilon = +10^{-2}$ level-4 collapse (solid), behind a MOTS from $t = 11$, and the unkicked level-4 throat (dotted), which inflates, horizonless, and grows in the monopole (counted to $t \approx 80$).

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